

ABCD ADHD menarche analysis: Output for cross-sectional models

Model 1 output

```
      Estimate Est.Error  Q2.5 Q97.5
R2      0.702    0.00322 0.696 0.708
```

```
Family: poisson
Links: mu = log
Formula: adhd_traits ~ menarche_status_p + age_sc + ethnrace + inr_sc + (1 | participant_id)
Data: analytic_df (Number of observations: 19381)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family_id (Number of levels: 4626)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.830	0.045	0.737	0.914	1.003	665	1190

~participant_id (Number of levels: 5297)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.042	0.035	0.974	1.110	1.002	835	2037

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.177	0.040	0.114	0.265	1.000	7021	6366

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.038	0.051	-0.061	0.136	1.000	7609	6068
menarche_status_pY	0.023	0.018	-0.013	0.058	1.000	16051	5994

age_sc	-0.040	0.016	-0.070	-0.009	1.000	14345	6338
ethnraceblack_nh	0.019	0.064	-0.108	0.145	1.000	11360	5496
ethnracehispanic	-0.048	0.060	-0.168	0.068	1.000	10992	6237
ethnraceother	0.055	0.069	-0.079	0.187	1.001	12117	6097
inr_sc	-0.084	0.023	-0.129	-0.041	1.001	18061	5754

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 1 effects expressed as % change in expected outcome

```
# A tibble: 1 x 4
  Variable Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>         <dbl>         <dbl>         <dbl>
1 Menarche Y      2.33          -1.29          5.99
```

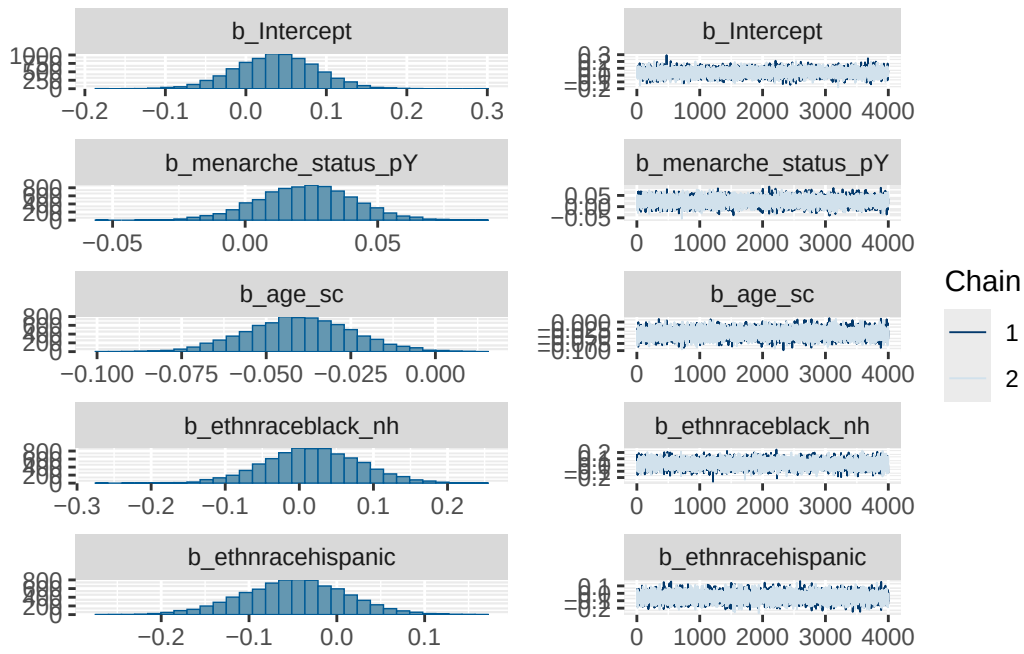
Model 1 diagnostics

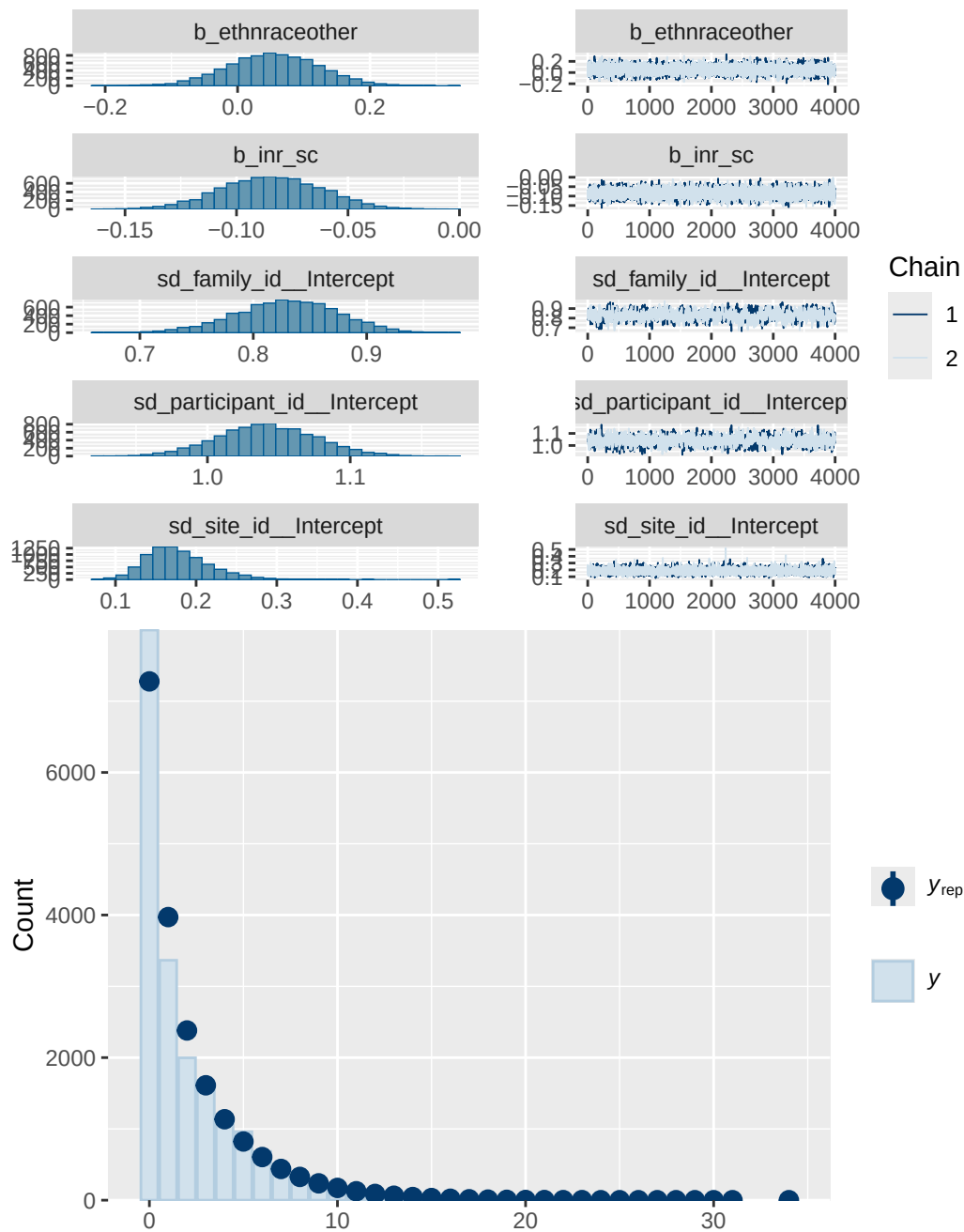
	prior	class	coef	group resp dpar
	normal(0, 1)	b		
	normal(0, 1)	b	age_sc	
	normal(0, 1)	b	ethnraceblack_nh	
	normal(0, 1)	b	ethnracehispanic	
	normal(0, 1)	b	ethnraceother	
	normal(0, 1)	b	inr_sc	
	normal(0, 1)	b	menarche_status_pY	
student_t(3, 0, 2.7)	Intercept			
student_t(3, 0, 2.7)	sd			
student_t(3, 0, 2.7)	sd			family_id
student_t(3, 0, 2.7)	sd	Intercept		family_id
student_t(3, 0, 2.7)	sd			participant_id
student_t(3, 0, 2.7)	sd	Intercept		participant_id
student_t(3, 0, 2.7)	sd			site_id
student_t(3, 0, 2.7)	sd	Intercept		site_id
nlpar lb ub tag	source			
	user			
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```

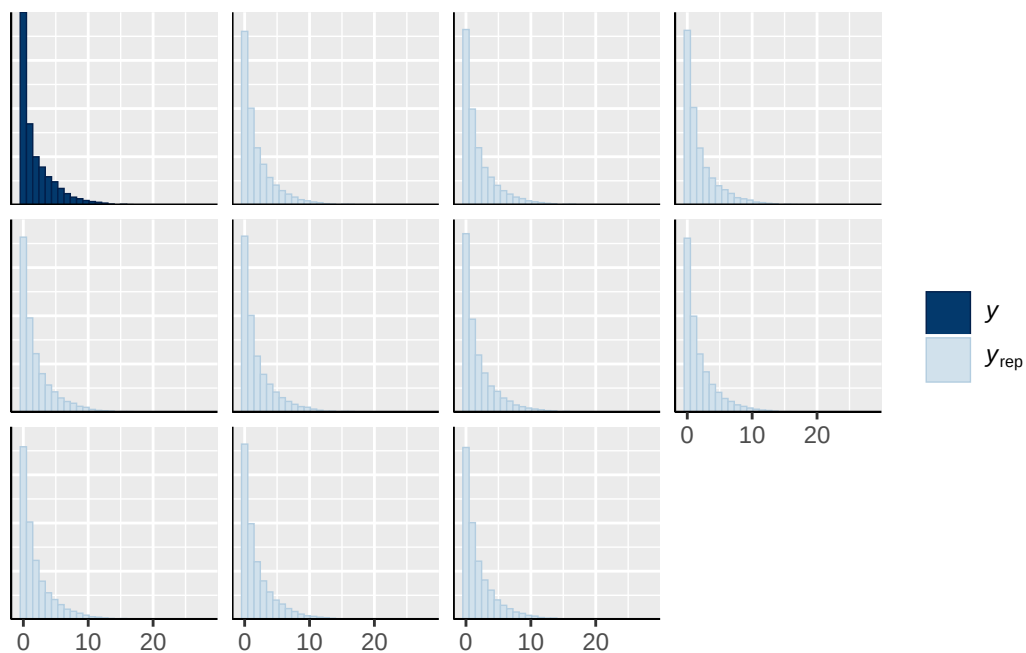
      (vectorized)
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0      default
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```

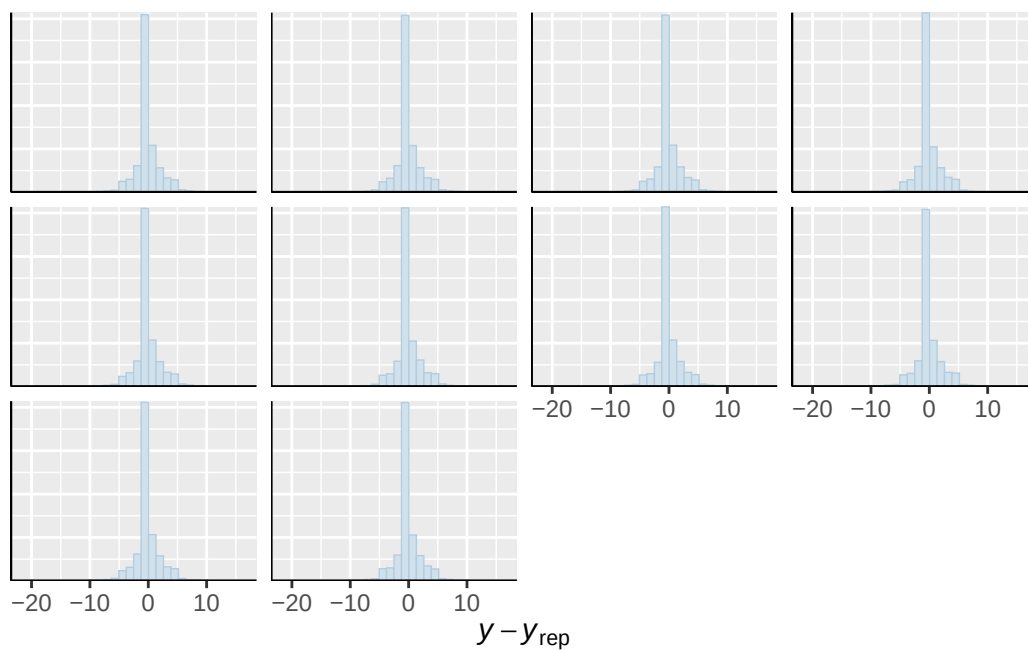




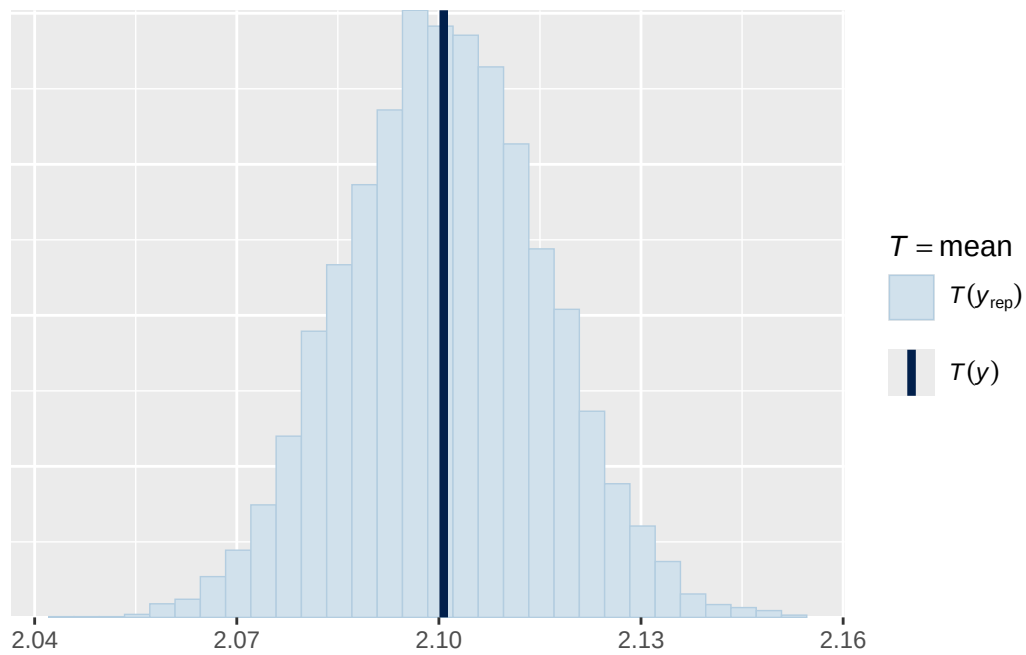
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



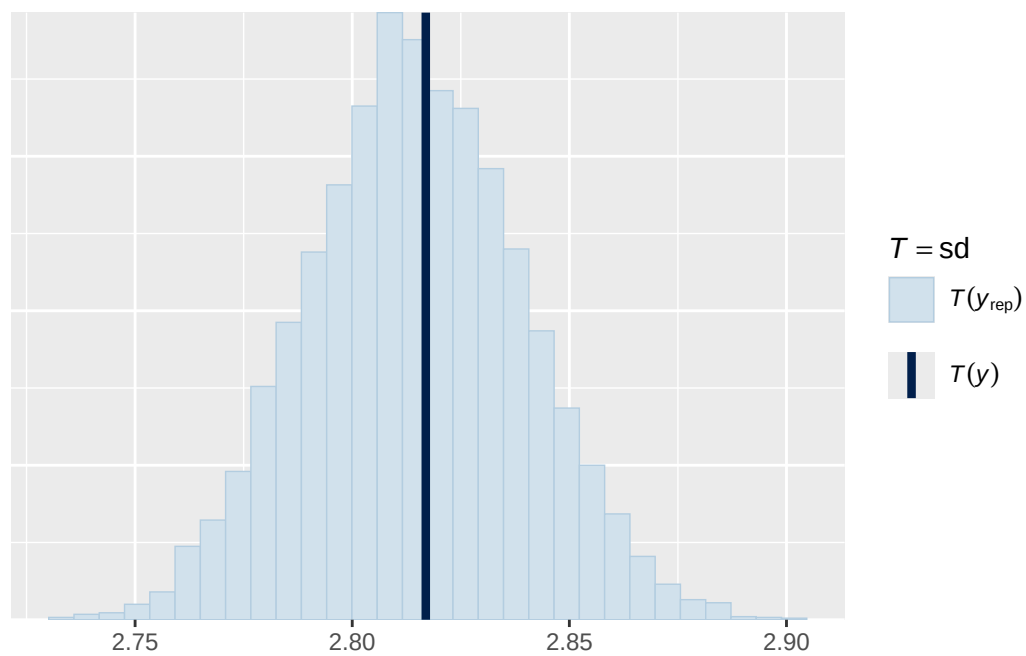
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

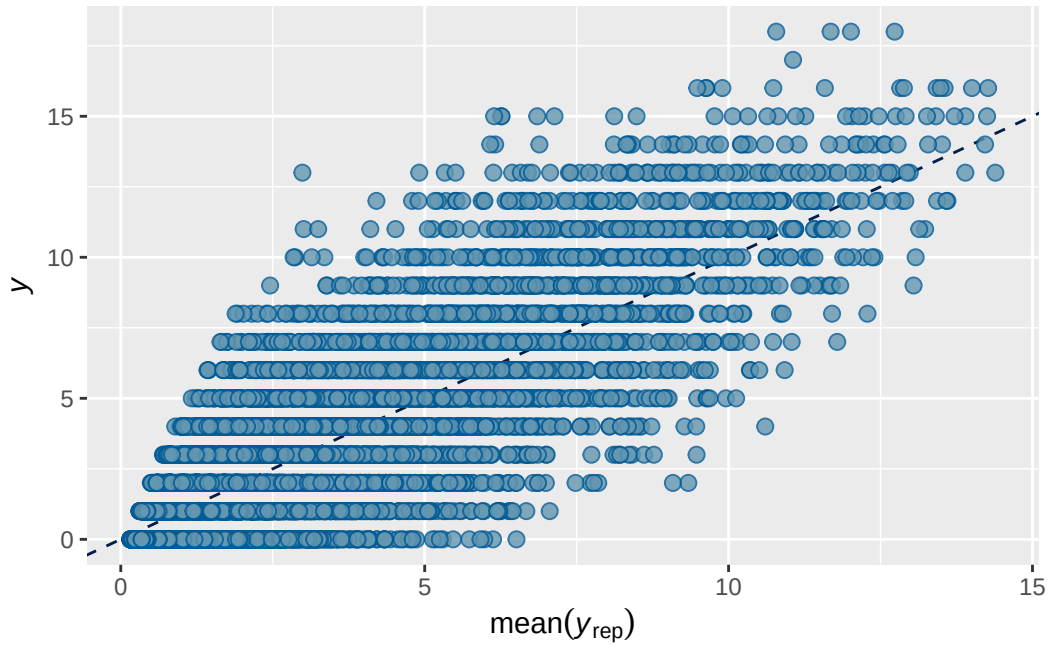


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Model 1b output

```

      Estimate Est.Error  Q2.5 Q97.5
R2    0.703    0.00329 0.696 0.709

```

Family: poisson

Links: mu = log

Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + (1 | participant_id)

Data: analytic_df (Number of observations: 19364)

Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 4619)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.834	0.046	0.740	0.920	1.005	584	1177

~participant_id (Number of levels: 5289)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.036	0.036	0.967	1.107	1.002	732	1751

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.176	0.039	0.111	0.264	1.000	6010	5359

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	0.044	0.051	-0.055	0.146	1.000	6379
breast_development_p_sc	0.059	0.019	0.023	0.095	1.001	12977
age_sc	-0.058	0.016	-0.089	-0.027	1.000	14206
ethnraceblack_nh	0.027	0.067	-0.105	0.156	1.000	8354
ethnracehispanic	-0.033	0.061	-0.154	0.087	1.000	9225
ethnraceother	0.048	0.067	-0.083	0.179	1.000	8624
inr_sc	-0.076	0.022	-0.119	-0.032	1.000	16951

	Tail_ESS
Intercept	5766
breast_development_p_sc	6378
age_sc	6212
ethnraceblack_nh	6155
ethnracehispanic	6030
ethnraceother	5726
inr_sc	5781

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 1b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

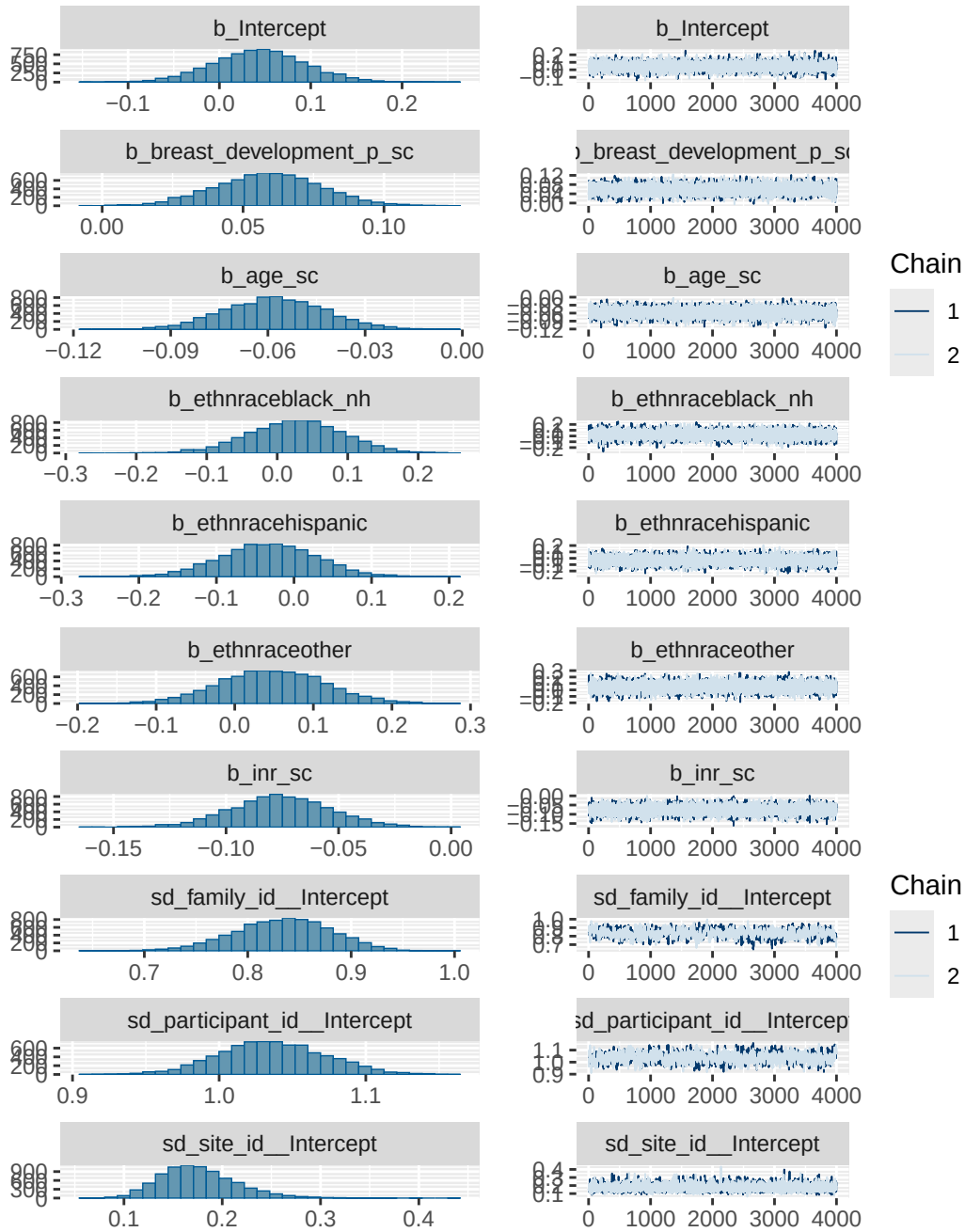
A tibble: 1 x 4

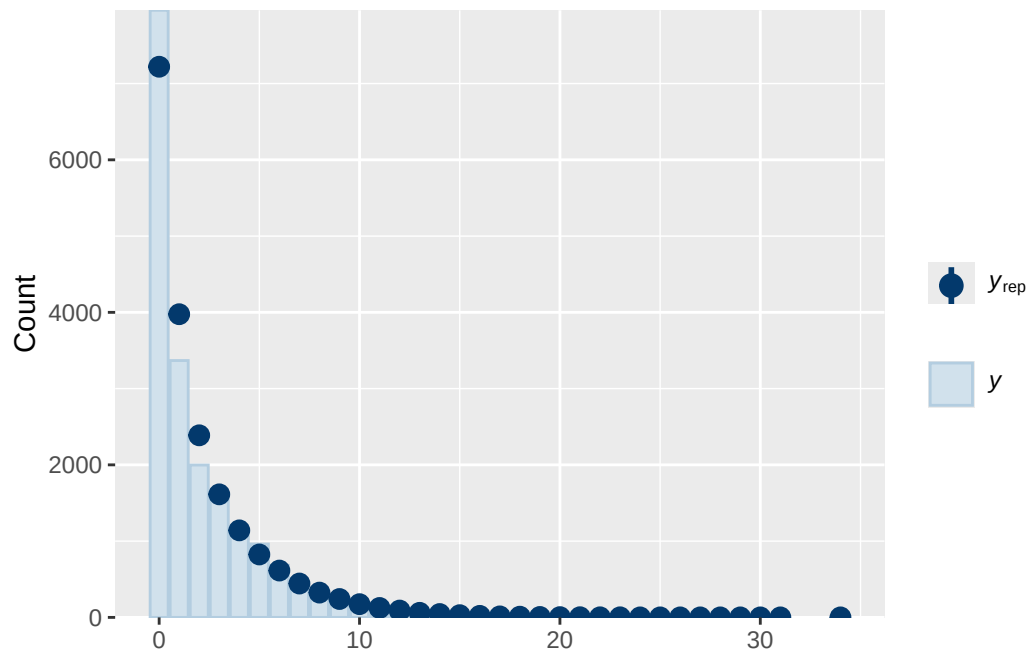
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development	3.7	1.43	6.05

Model 1b diagnostics

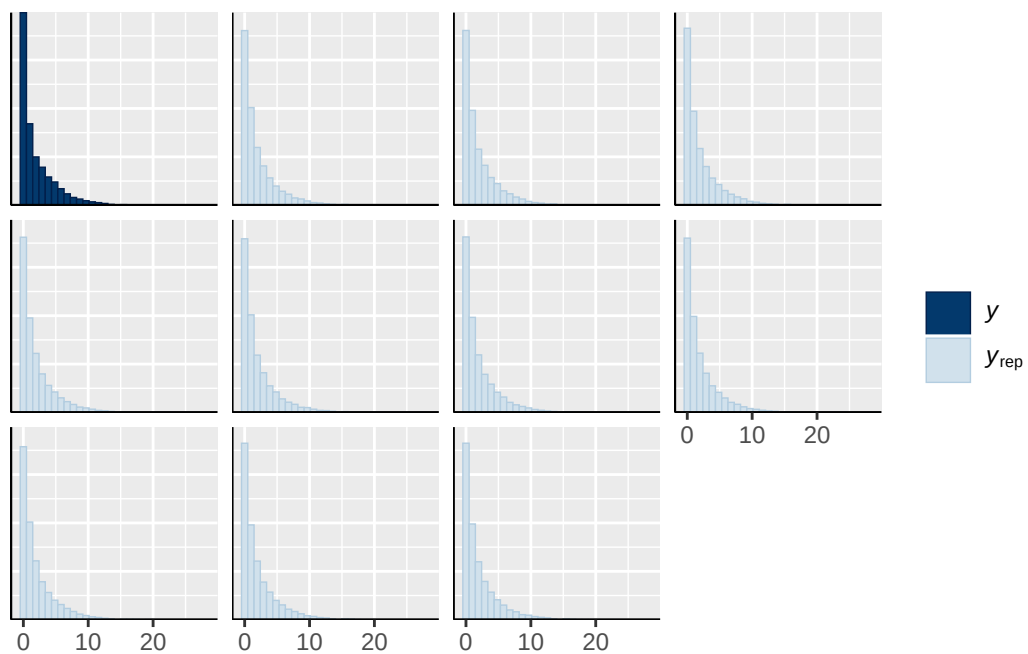
prior	class	coef	group resp
normal(0, 1)	b		
normal(0, 1)	b	age_sc	
normal(0, 1)	b	breast_development_p_sc	
normal(0, 1)	b	ethnraceblack_nh	
normal(0, 1)	b	ethnracehispanic	

normal(0, 1)	b	ethn	race	other
normal(0, 1)	b		inr_sc	
student_t(3, 0, 2.7)	Intercept			
student_t(3, 0, 2.7)	sd			
student_t(3, 0, 2.7)	sd			family_id
student_t(3, 0, 2.7)	sd	Intercept		family_id
student_t(3, 0, 2.7)	sd			participant_id
student_t(3, 0, 2.7)	sd	Intercept		participant_id
student_t(3, 0, 2.7)	sd			site_id
student_t(3, 0, 2.7)	sd	Intercept		site_id
dpar nlpar lb ub tag	source			
	user			
	(vectorized)			
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	(vectorized)			
	(vectorized)			
	default			
0	default			
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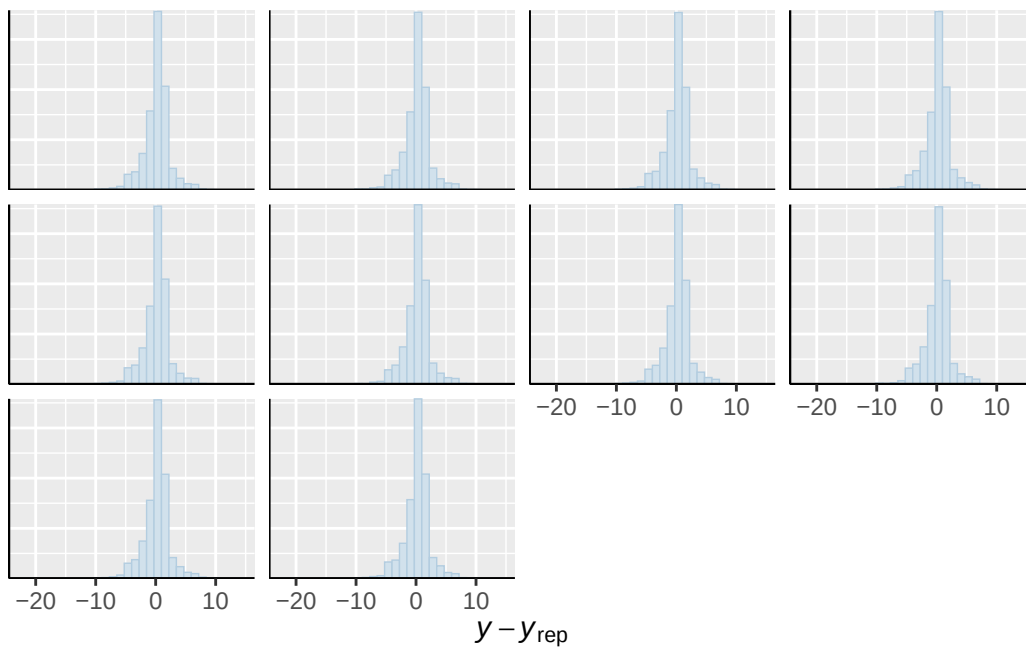




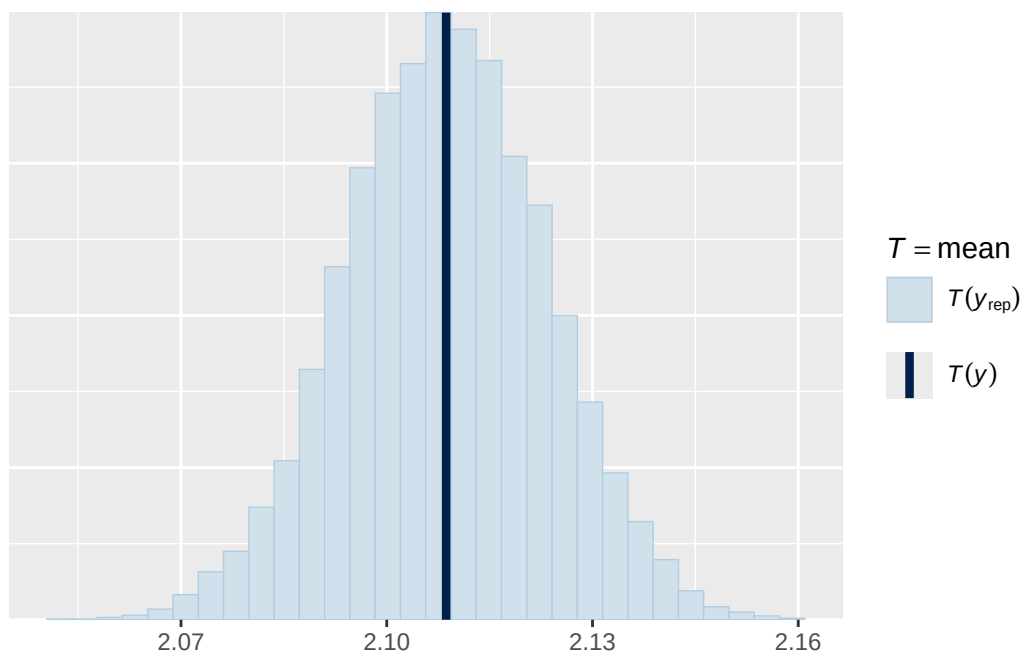
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



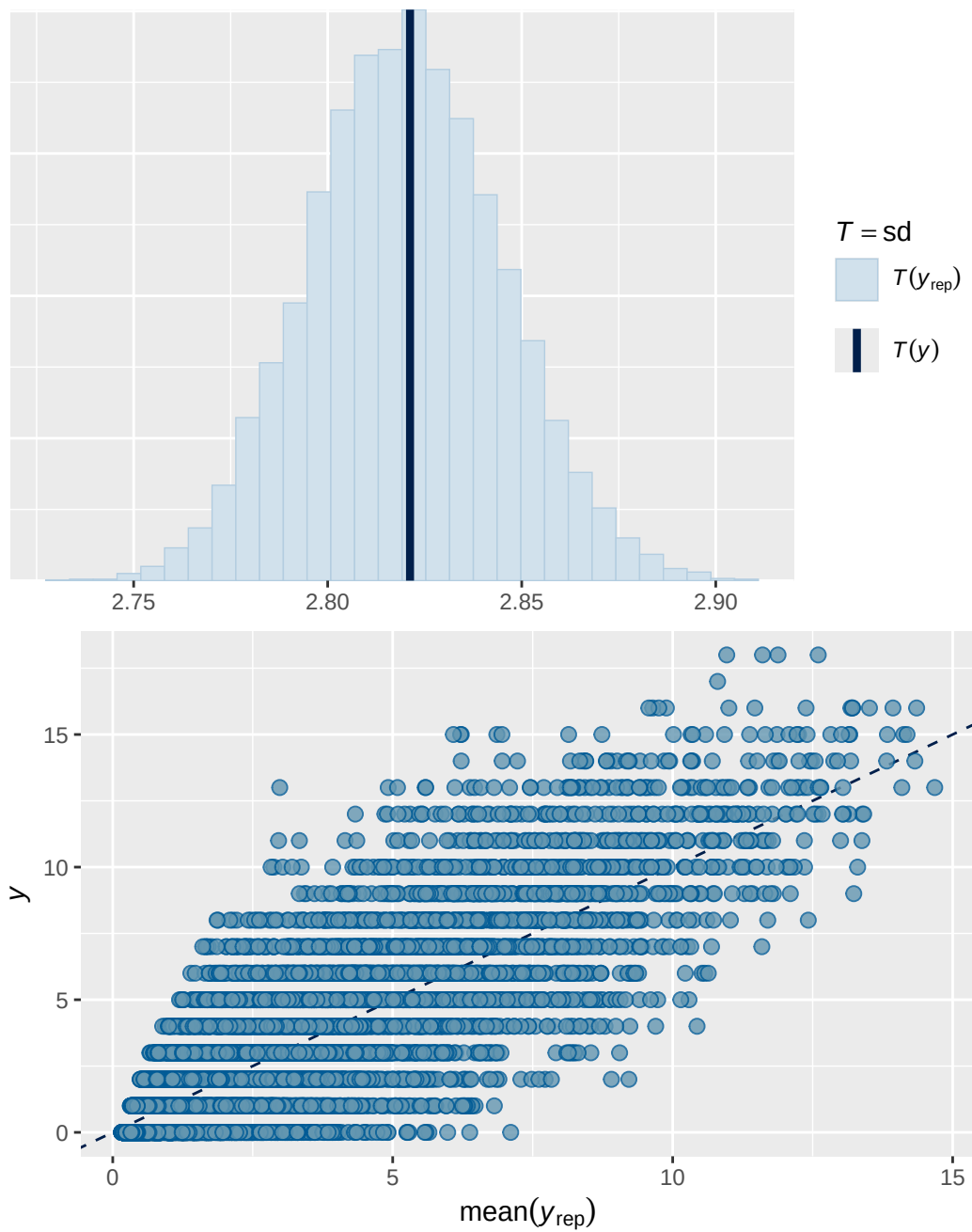
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 2 output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.839	0.00252	0.833	0.843

Family: poisson
 Links: mu = log
 Formula: internalising ~ menarche_status_p * adhd_status_life + age_sc + ethnrace + inr_sc +
 Data: analytic_df (Number of observations: 19375)
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 4620)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.761	0.018	0.726	0.795	1.002	926	1709

~obs_id (Number of levels: 19375)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.447	0.006	0.434	0.459	1.002	1555	3061

~participant_id (Number of levels: 5291)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.450	0.022	0.408	0.494	1.002	491	1018

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.119	0.027	0.074	0.181	1.001	3041	2635

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat
Intercept	1.069	0.035	0.999	1.138	1.001
menarche_status_pY	0.085	0.019	0.048	0.120	1.001
adhd_status_lifeY	0.630	0.035	0.560	0.700	1.001
age_sc	0.053	0.015	0.023	0.083	1.000
ethnraceblack_nh	-0.401	0.046	-0.490	-0.311	1.000
ethnracehispanic	-0.070	0.042	-0.151	0.013	1.000
ethnraceother	-0.018	0.045	-0.106	0.073	1.001
inr_sc	-0.048	0.021	-0.089	-0.008	1.000
menarche_status_pY:adhd_status_lifeY	-0.020	0.031	-0.079	0.040	1.000

	Bulk_ESS	Tail_ESS
Intercept	2999	2686
menarche_status_pY	4768	3184
adhd_status_lifeY	3793	3257
age_sc	5096	3064
ethnraceblack_nh	4020	3029
ethnracehispanic	3863	2661
ethnraceother	4413	3143

inr_sc	5848	3116
menarche_status_pY:adhd_status_lifeY	5492	3315

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 2 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

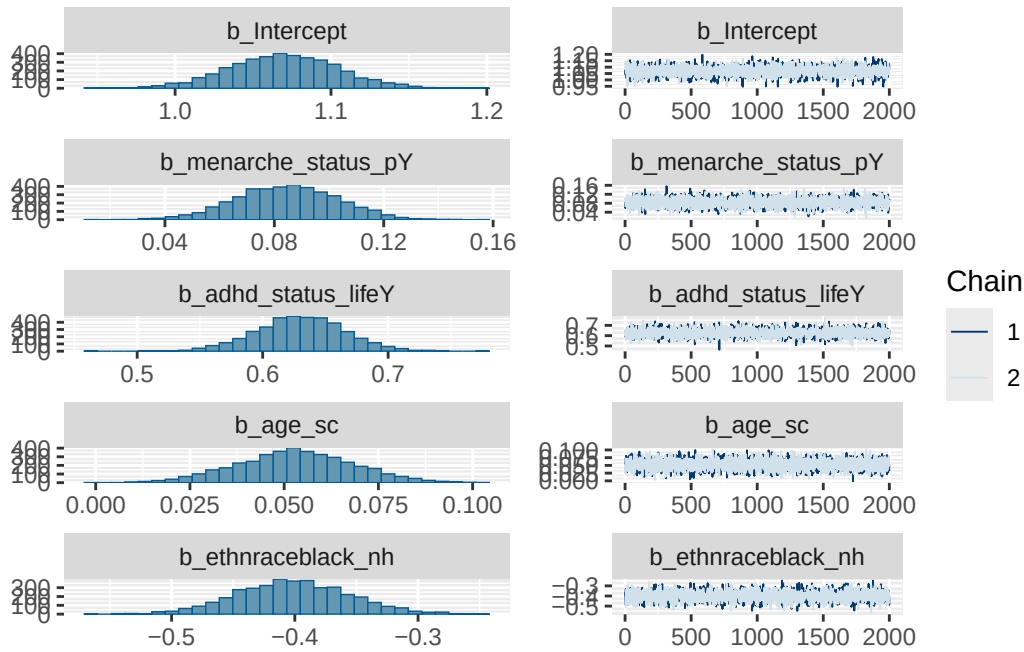
A tibble: 3 x 4

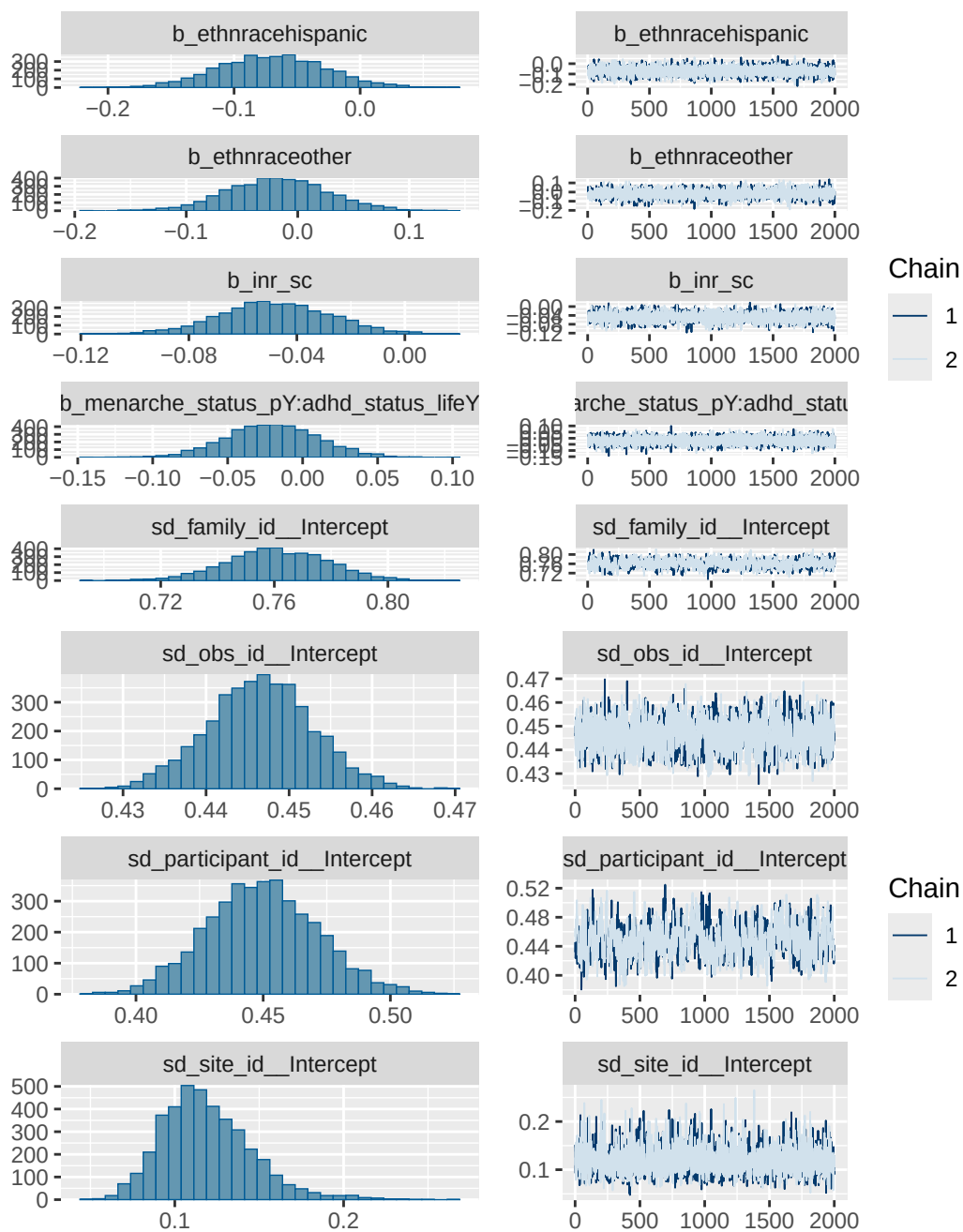
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Menarche (yes vs no) [ADHD = 0]	8.82	4.92	12.7
2 ADHD (yes vs no) [Menarche = 0]	87.8	75.1	101.
3 Interaction (Menarche × ADHD)	-2	-7.6	4.06

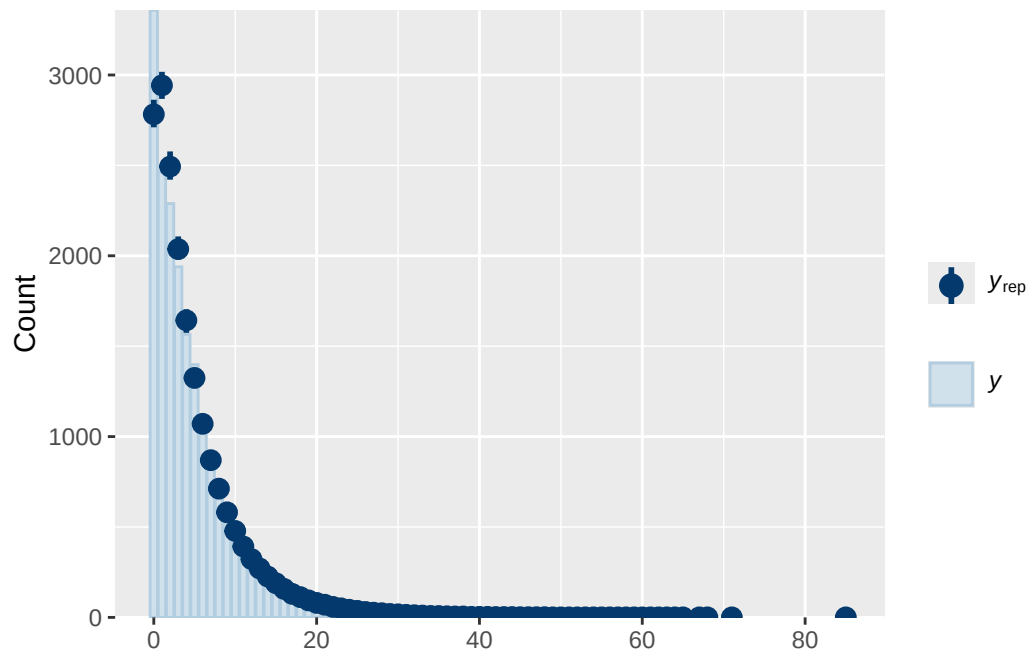
Model 2 diagnostics

prior	class	coef
normal(0, 1)	b	
normal(0, 1)	b	adhd_status_lifeY
normal(0, 1)	b	age_sc
normal(0, 1)	b	ethnraceblack_nh
normal(0, 1)	b	ethnracehispanic
normal(0, 1)	b	ethnraceother
normal(0, 1)	b	inr_sc
normal(0, 1)	b	menarche_status_pY
normal(0, 1)	b	menarche_status_pY:adhd_status_lifeY
student_t(3, 1.1, 2.5)	Intercept	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
group resp dpar nlpar lb ub tag		source
		user

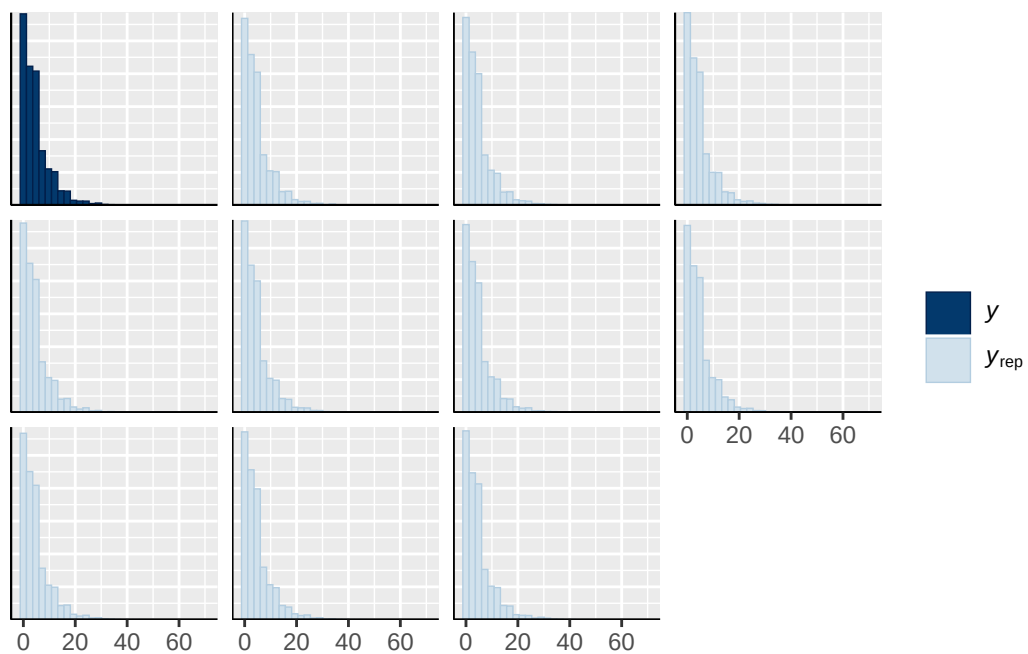
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	default	
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
obs_id	0	(vectorized)
obs_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



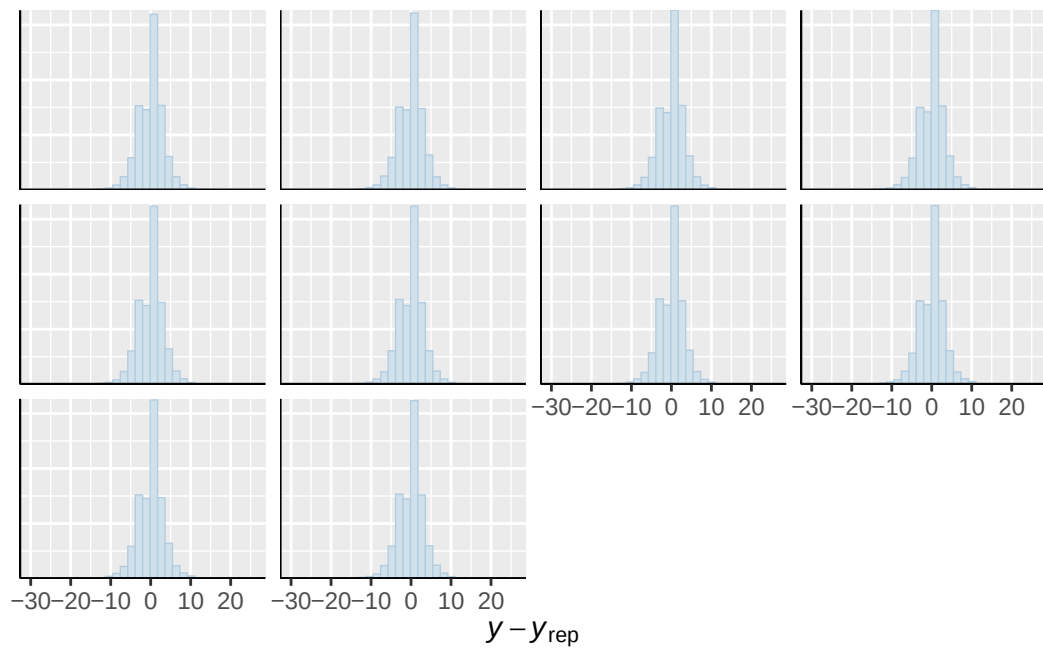




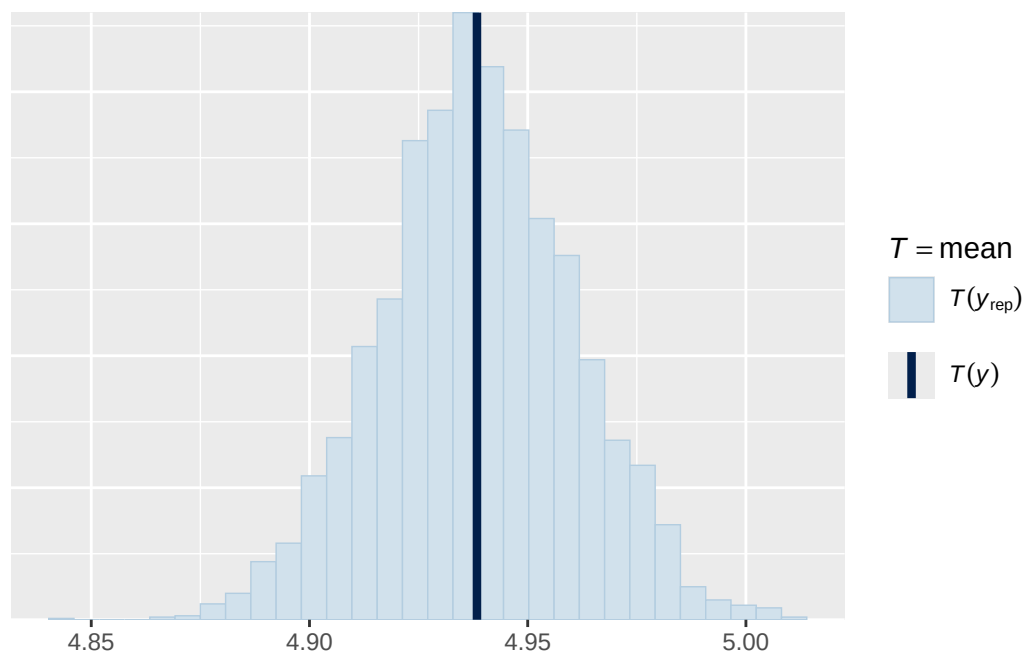
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



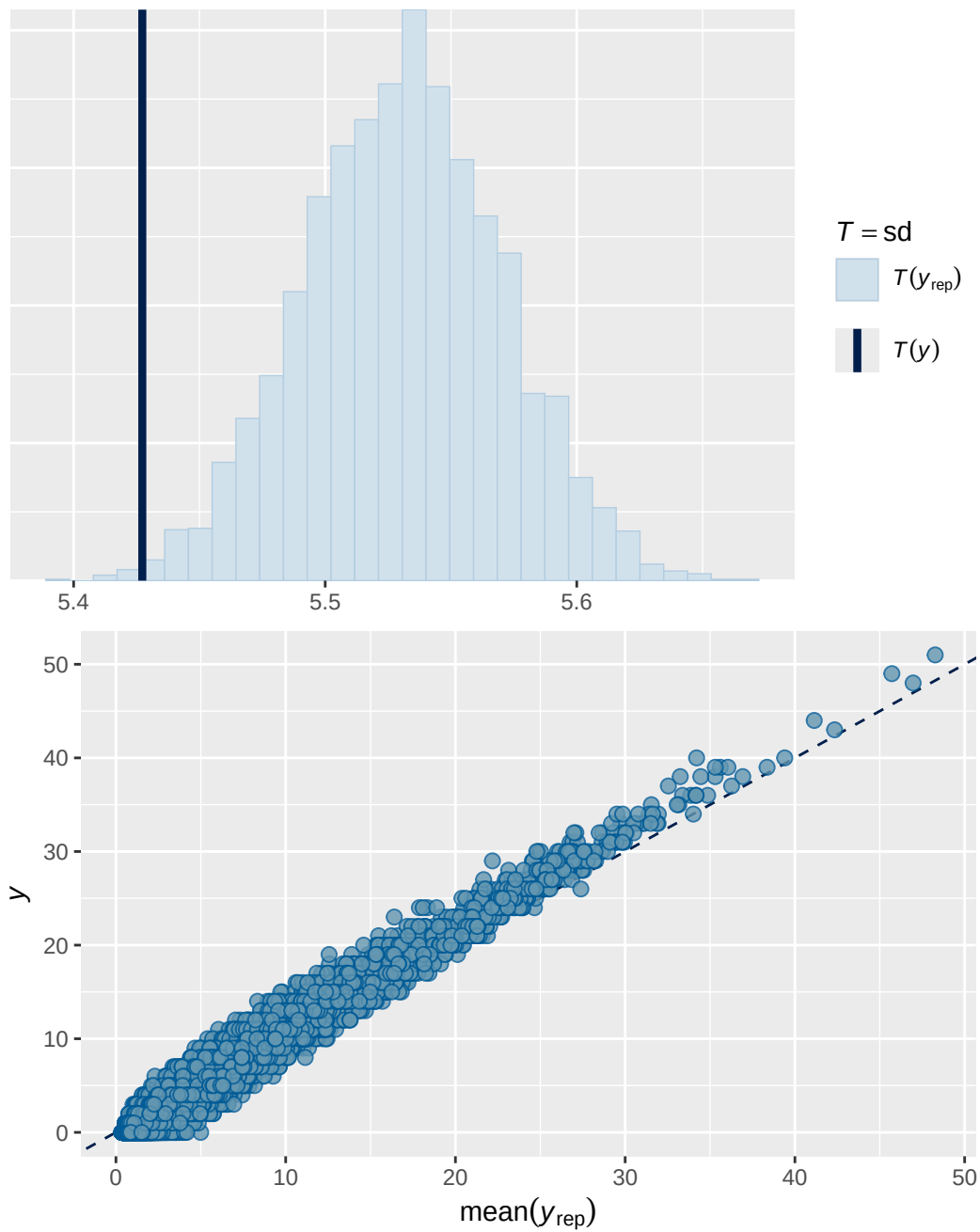
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 2b output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.839	0.00252	0.834	0.844

Family: poisson
 Links: mu = log
 Formula: internalising ~ breast_development_p_sc * adhd_status_life + age_sc + ethnrace + inr_sc
 Data: analytic_df (Number of observations: 19358)
 Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
 total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 4613)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.760	0.018	0.724	0.794	1.002	1121	1823

~obs_id (Number of levels: 19358)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.446	0.006	0.434	0.459	1.000	3082	4811

~participant_id (Number of levels: 5283)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.450	0.022	0.407	0.493	1.002	647	1089

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.119	0.026	0.077	0.179	1.000	3098	4125

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	1.094	0.034	1.025	1.164
breast_development_p_sc	0.085	0.020	0.047	0.123
adhd_status_lifeY	0.623	0.033	0.557	0.688
age_sc	0.058	0.015	0.028	0.088
ethnraceblack_nh	-0.393	0.045	-0.483	-
ethnracehispanic	-0.059	0.041	-0.140	0.020
ethnraceother	-0.018	0.046	-0.106	0.072
inr_sc	-0.042	0.020	-0.081	-
breast_development_p_sc:adhd_status_lifeY	-0.037	0.033	-0.103	0.026

	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.001	2867	3494
breast_development_p_sc	1.001	4431	4816
adhd_status_lifeY	1.000	3609	4548
age_sc	1.000	4709	4684

ethnraceblack_nh	1.001	2975	4358
ethnracehispanic	1.001	3858	4223
ethnraceother	1.001	3268	4697
inr_sc	1.001	4781	5121
breast_development_p_sc:adhd_status_lifeY	1.001	5083	5251

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 2b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

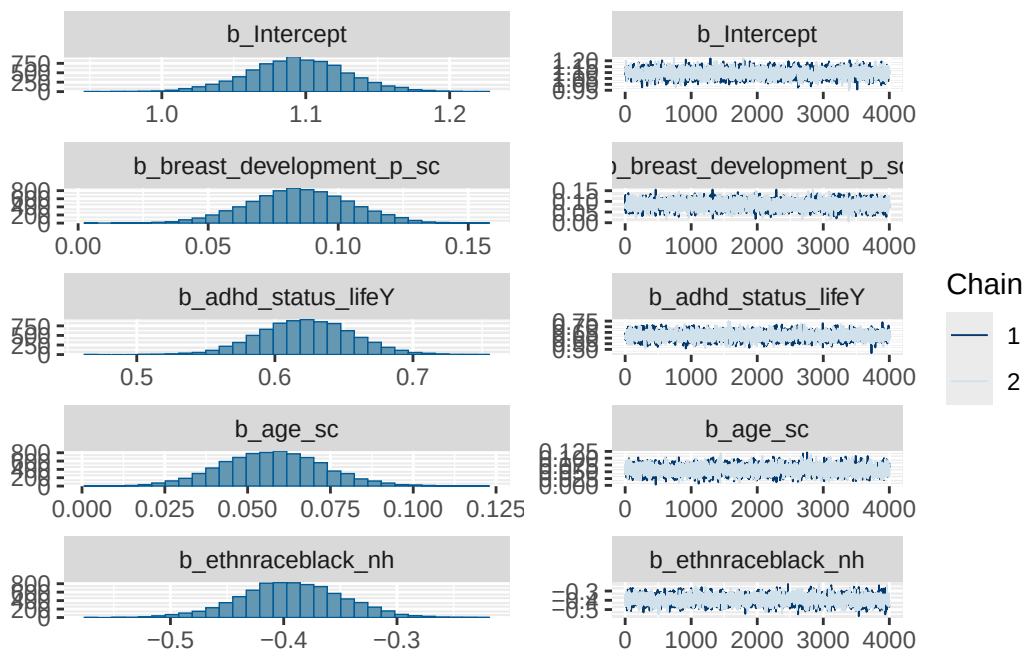
A tibble: 3 x 4

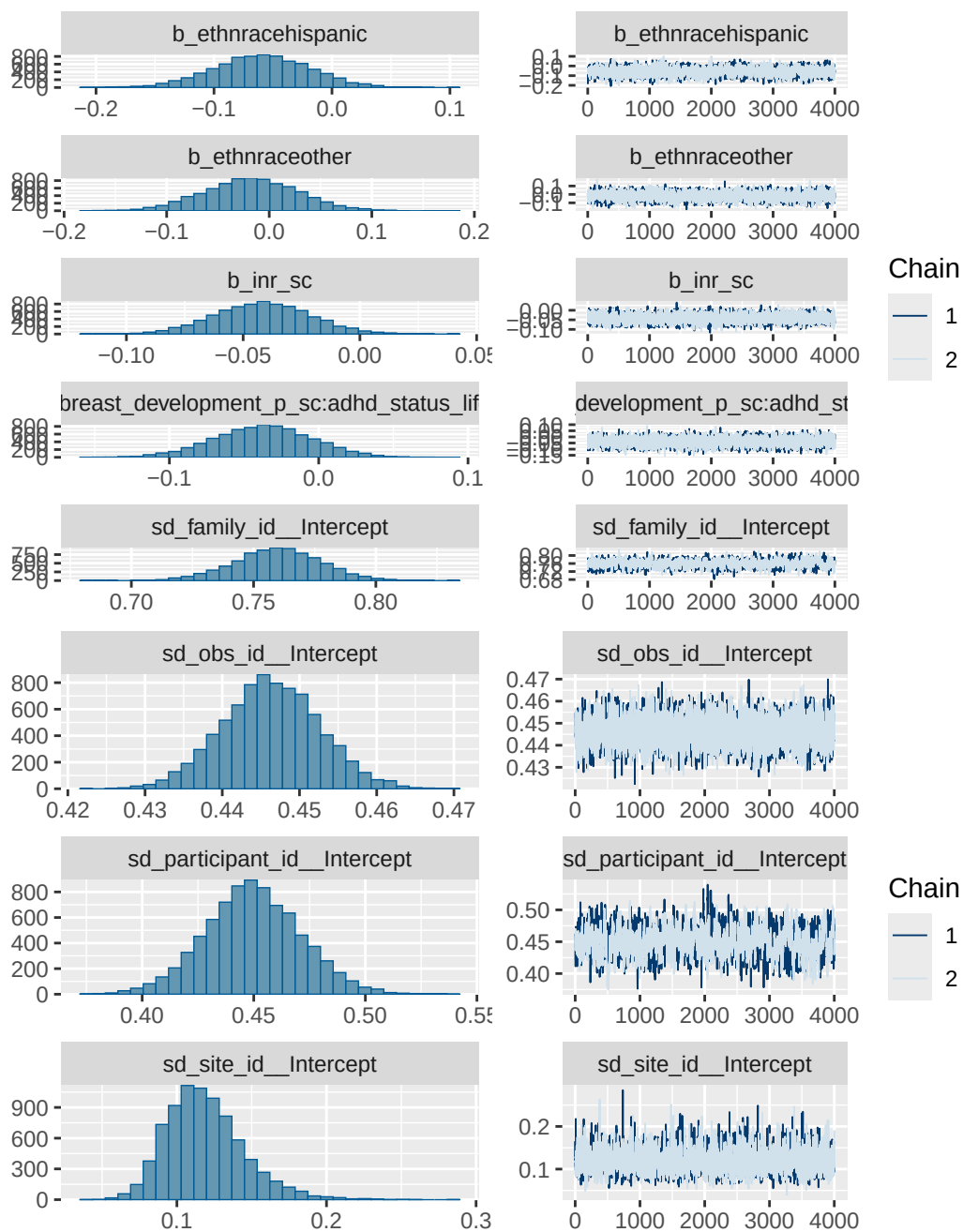
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development (per 1-unit) [ADHD ~	5.39	2.91	7.89
2 ADHD (yes vs no) [Breast development =~	86.4	74.6	98.9
3 Interaction (Breast development × ADHD)	-2.26	-6.15	1.63

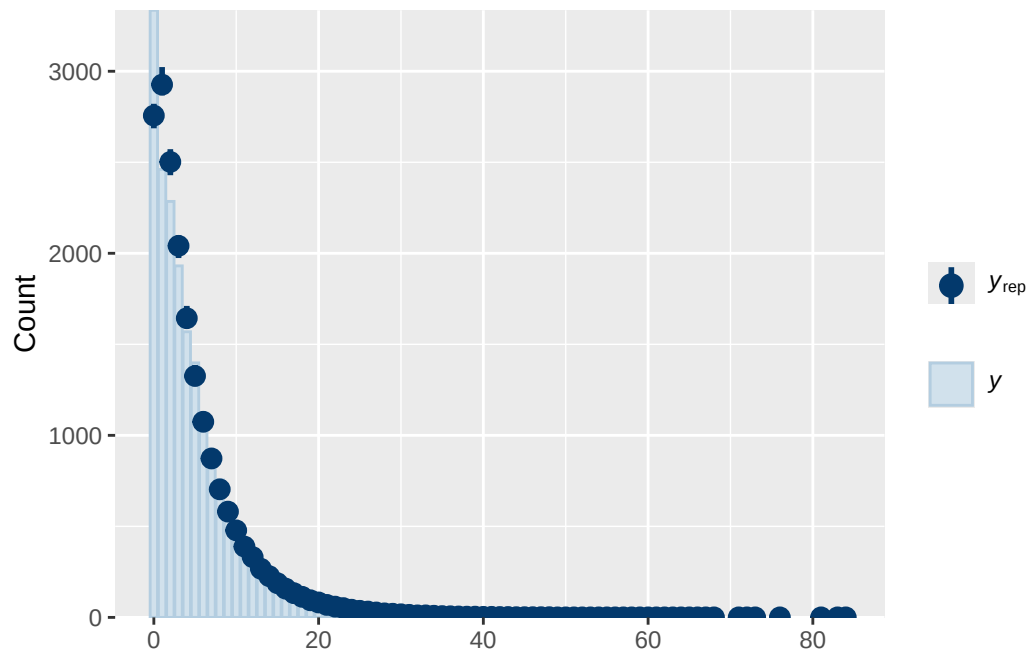
Model 2b diagnostics

	prior	class	coef
	normal(0, 1)	b	
	normal(0, 1)	b	adhd_status_lifeY
	normal(0, 1)	b	age_sc
	normal(0, 1)	b	breast_development_p_sc
	normal(0, 1)	b	breast_development_p_sc:adhd_status_lifeY
	normal(0, 1)	b	ethnraceblack_nh
	normal(0, 1)	b	ethnracehispanic
	normal(0, 1)	b	ethnraceother
	normal(0, 1)	b	inr_sc
student_t(3, 1.1, 2.5)	Intercept		
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept

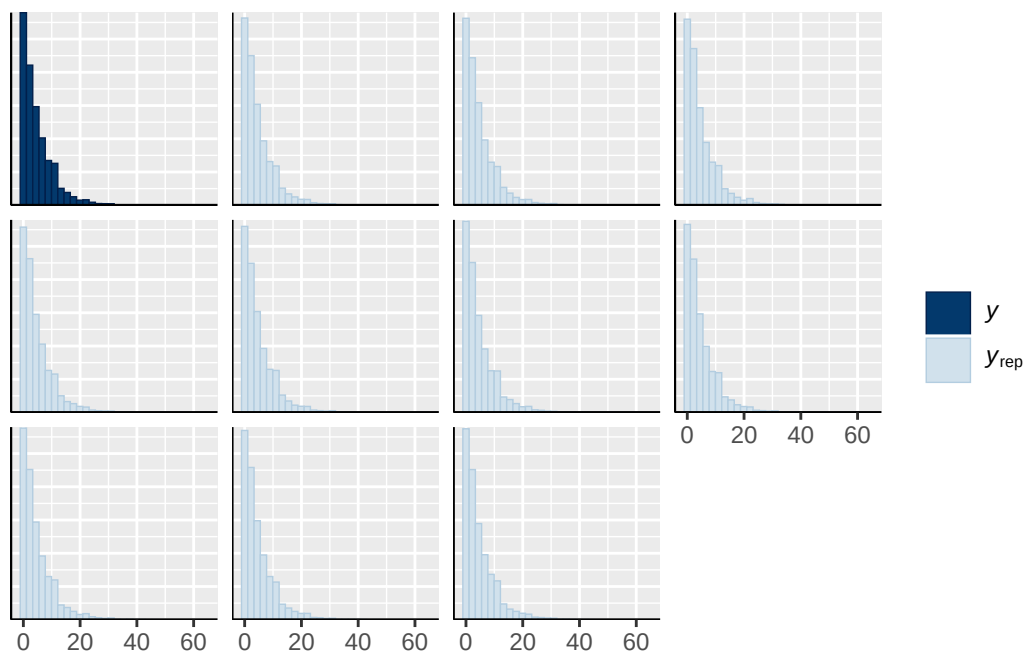
student_t(3, 0, 2.5)	sd	Intercept
group resp dpar nlpar lb ub tag		source
		user
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
obs_id	0	(vectorized)
obs_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



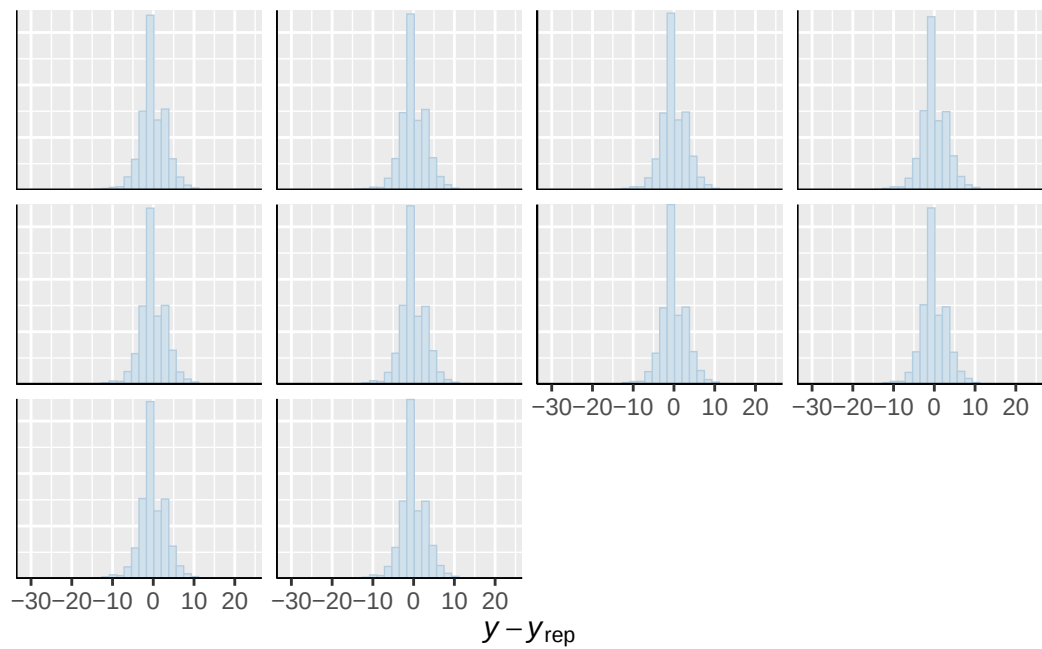




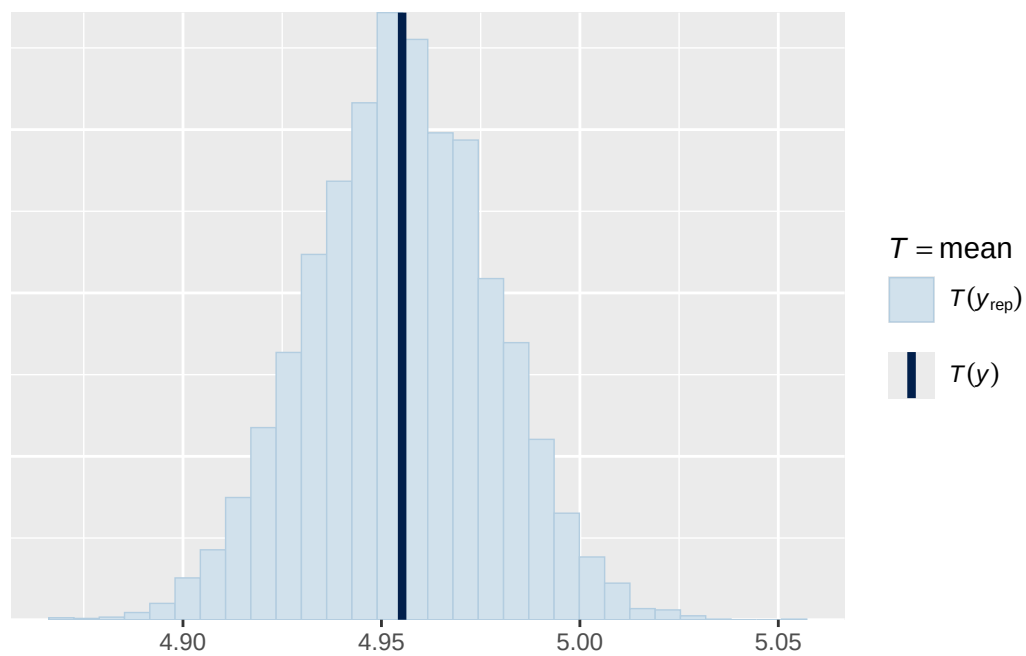
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



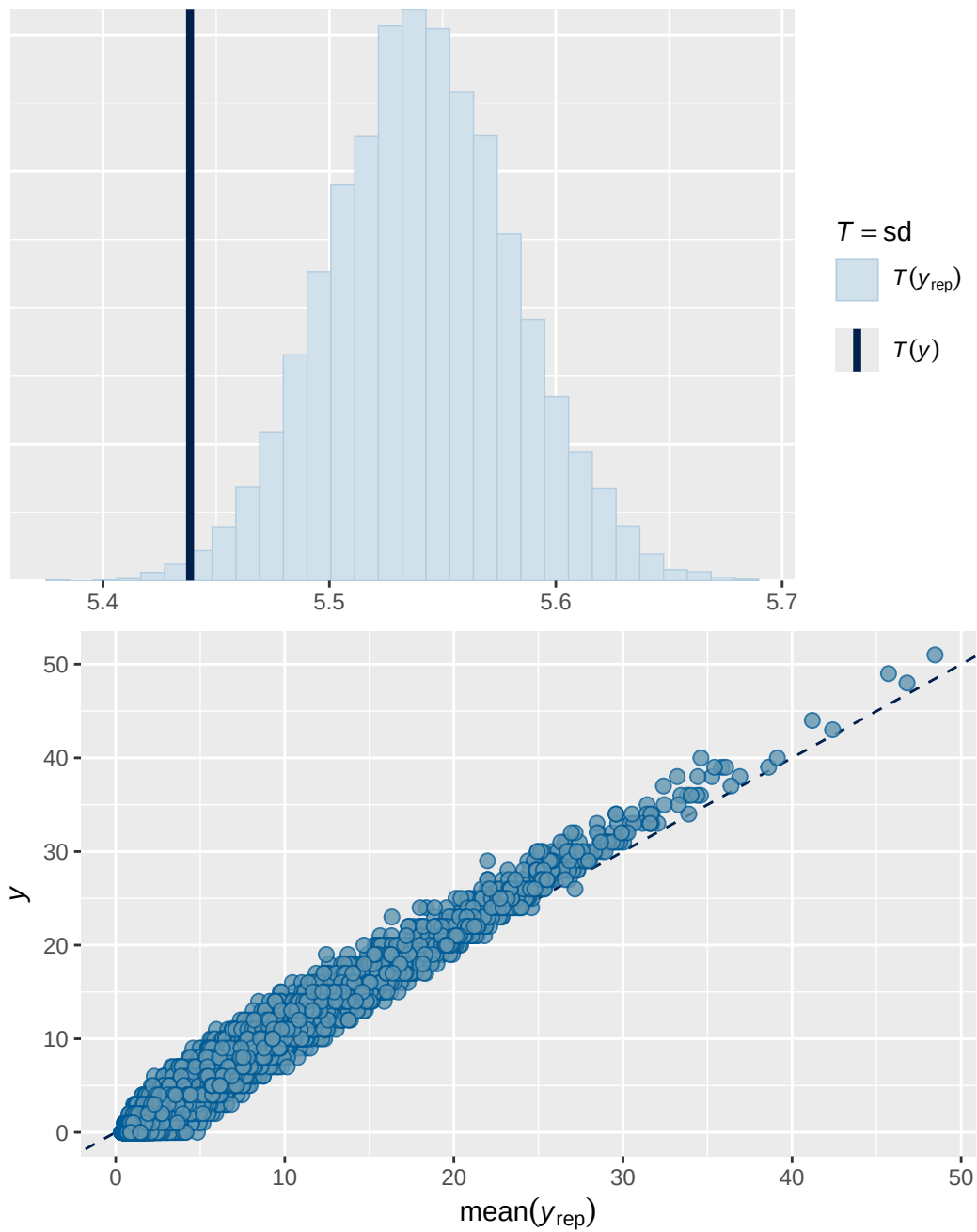
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 3 output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.859	0.00254	0.854	0.864

Family: poisson
 Links: mu = log
 Formula: externalising ~ menarche_status_p * adhd_status_life + age_sc + ethnrace + inr_sc +
 Data: analytic_df (Number of observations: 19375)
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 4620)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.866	0.031	0.804	0.925	1.001	571	981

~obs_id (Number of levels: 19375)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.497	0.008	0.481	0.513	1.001	1635	2749

~participant_id (Number of levels: 5291)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.767	0.030	0.708	0.827	1.001	482	929

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.154	0.034	0.097	0.229	1.000	2806	2777

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat
Intercept	0.117	0.046	0.029	0.208	1.000
menarche_status_pY	0.104	0.024	0.056	0.150	1.000
adhd_status_lifeY	1.337	0.046	1.250	1.428	1.000
age_sc	-0.133	0.020	-0.171	-0.095	1.000
ethnraceblack_nh	-0.005	0.058	-0.117	0.108	1.000
ethnracehispanic	-0.011	0.054	-0.118	0.096	1.000
ethnraceother	-0.008	0.060	-0.126	0.108	1.000
inr_sc	-0.185	0.026	-0.237	-0.133	1.000
menarche_status_pY:adhd_status_lifeY	-0.038	0.037	-0.110	0.035	1.000

	Bulk_ESS	Tail_ESS
Intercept	2834	2840
menarche_status_pY	4417	2993
adhd_status_lifeY	3473	2959
age_sc	4313	3393
ethnraceblack_nh	3811	3114
ethnracehispanic	3719	3092
ethnraceother	4016	3057

inr_sc	4859	2898
menarche_status_pY:adhd_status_lifeY	4283	2964

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

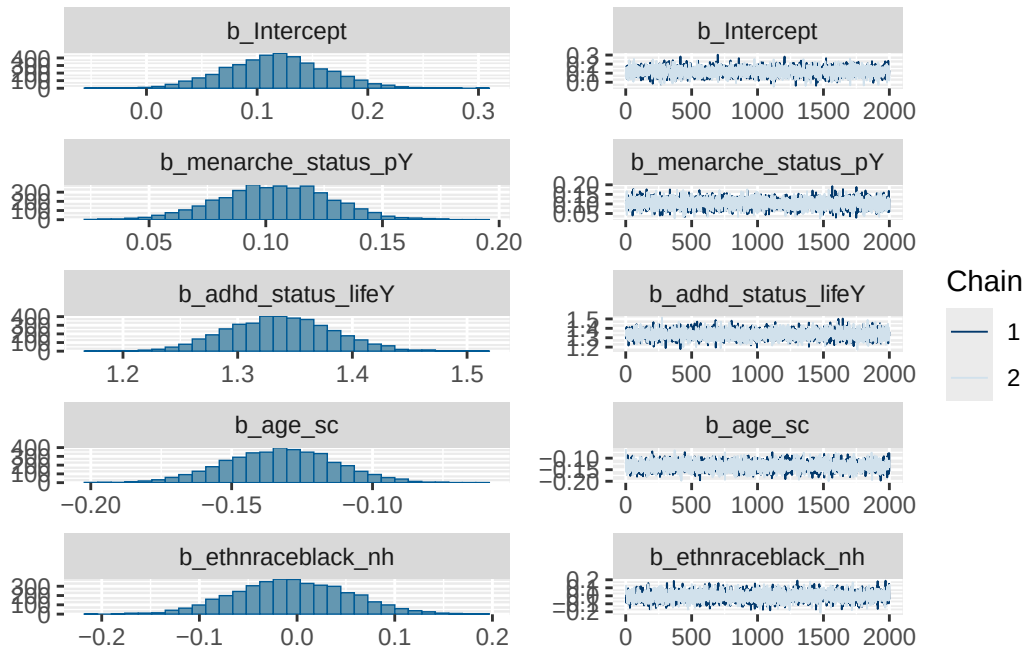
Model 3 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

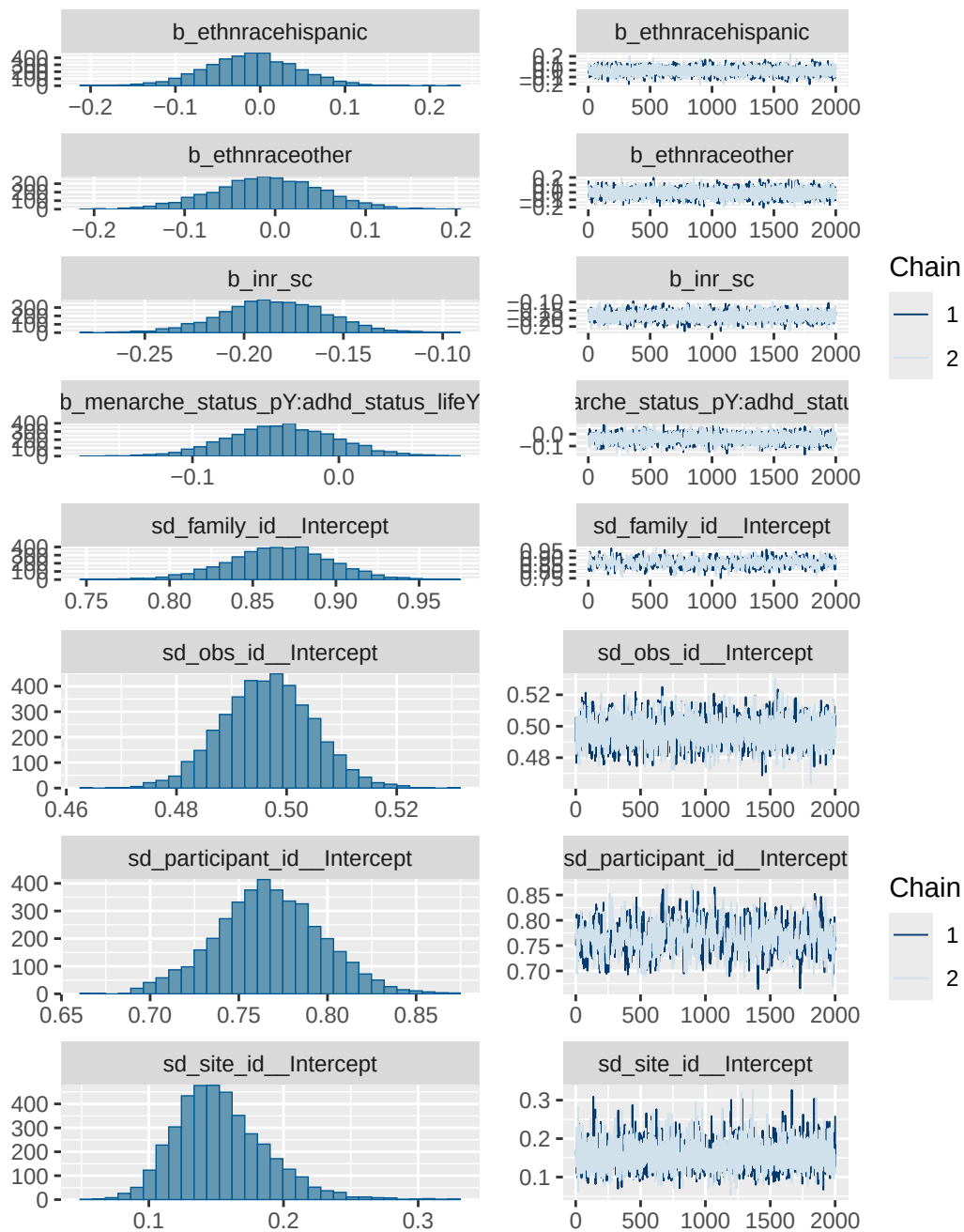
```
# A tibble: 3 x 4
  Variable                                Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                                <dbl>         <dbl>         <dbl>
1 Menarche (yes vs no) [ADHD = 0]      11.0          5.79          16.2
2 ADHD (yes vs no) [Menarche = 0]    281.          249.          317.
3 Interaction (Menarche × ADHD)      -3.78        -10.4          3.56
```

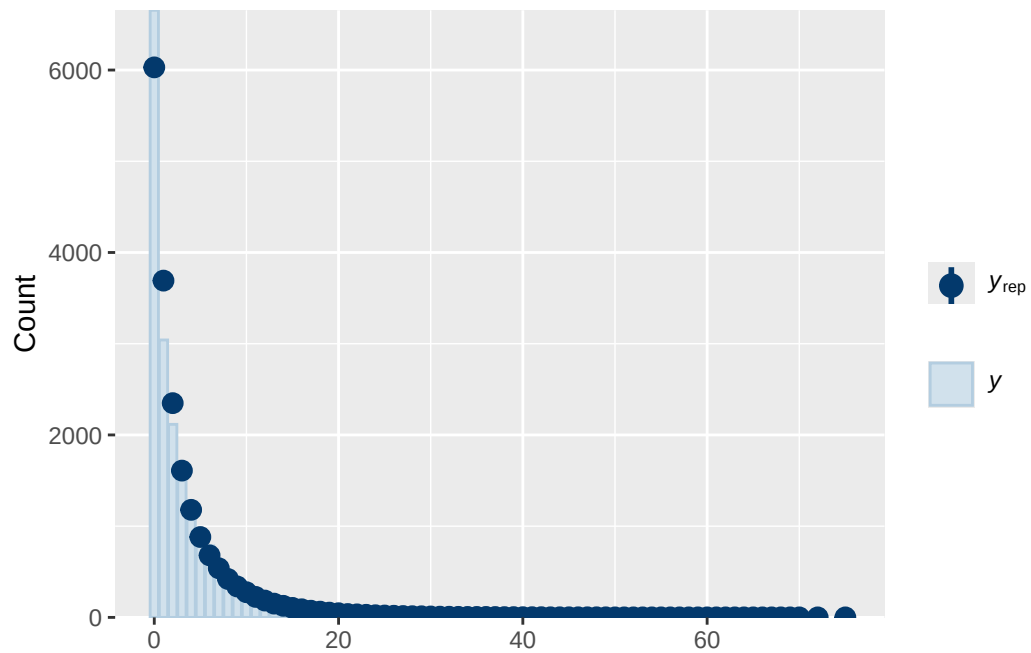
Model 3 diagnostics

	prior	class		coef
	normal(0, 1)	b		
	normal(0, 1)	b	adhd_status_lifeY	
	normal(0, 1)	b	age_sc	
	normal(0, 1)	b	ethnraceblack_nh	
	normal(0, 1)	b	ethnracehispanic	
	normal(0, 1)	b	ethnraceother	
	normal(0, 1)	b	inr_sc	
	normal(0, 1)	b	menarche_status_pY	
	normal(0, 1)	b	menarche_status_pY:adhd_status_lifeY	
student_t(3, 0, 2.9)	Intercept			
student_t(3, 0, 2.9)	sd			
student_t(3, 0, 2.9)	sd			
student_t(3, 0, 2.9)	sd		Intercept	
student_t(3, 0, 2.9)	sd			
student_t(3, 0, 2.9)	sd		Intercept	
student_t(3, 0, 2.9)	sd			
student_t(3, 0, 2.9)	sd		Intercept	
student_t(3, 0, 2.9)	sd			
student_t(3, 0, 2.9)	sd		Intercept	
	group resp dpar nlpar lb ub tag		source	
			user	

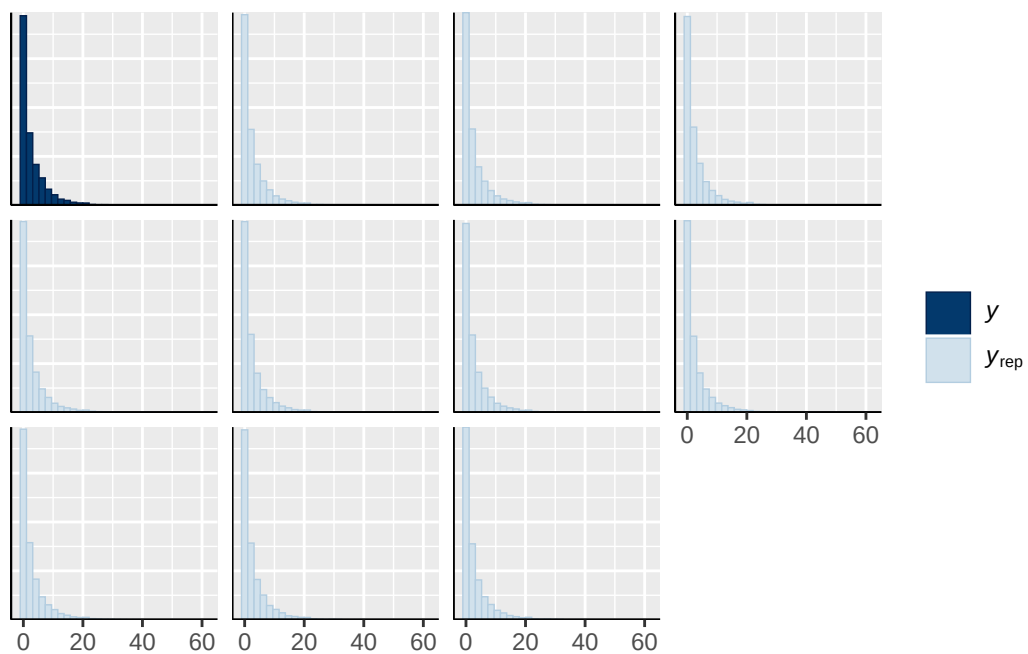
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family_id	0	(vectorized)
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obs_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



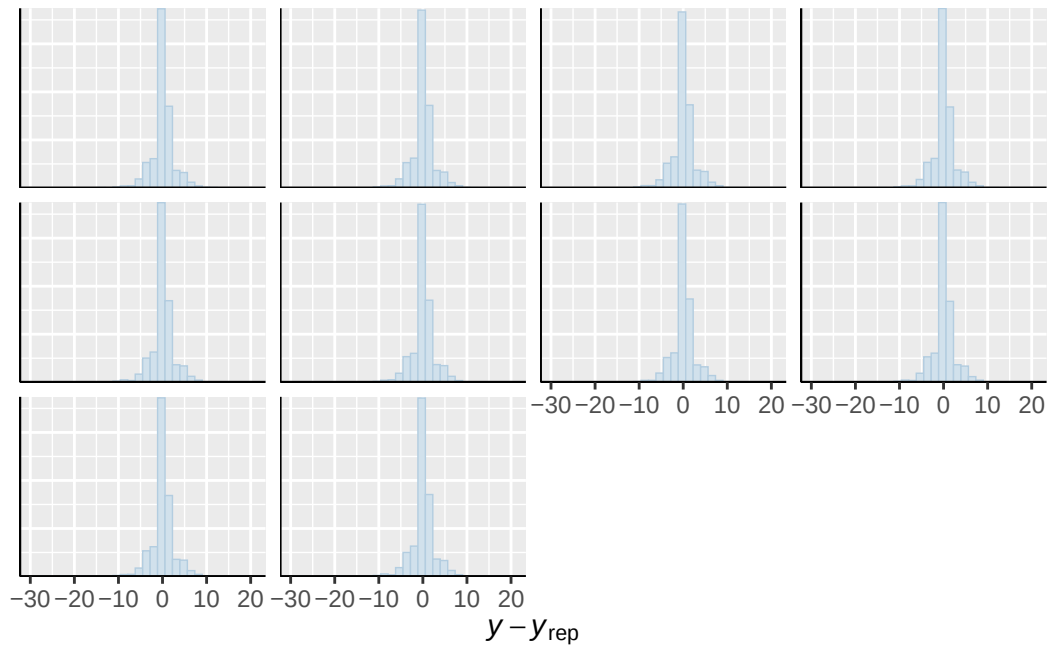




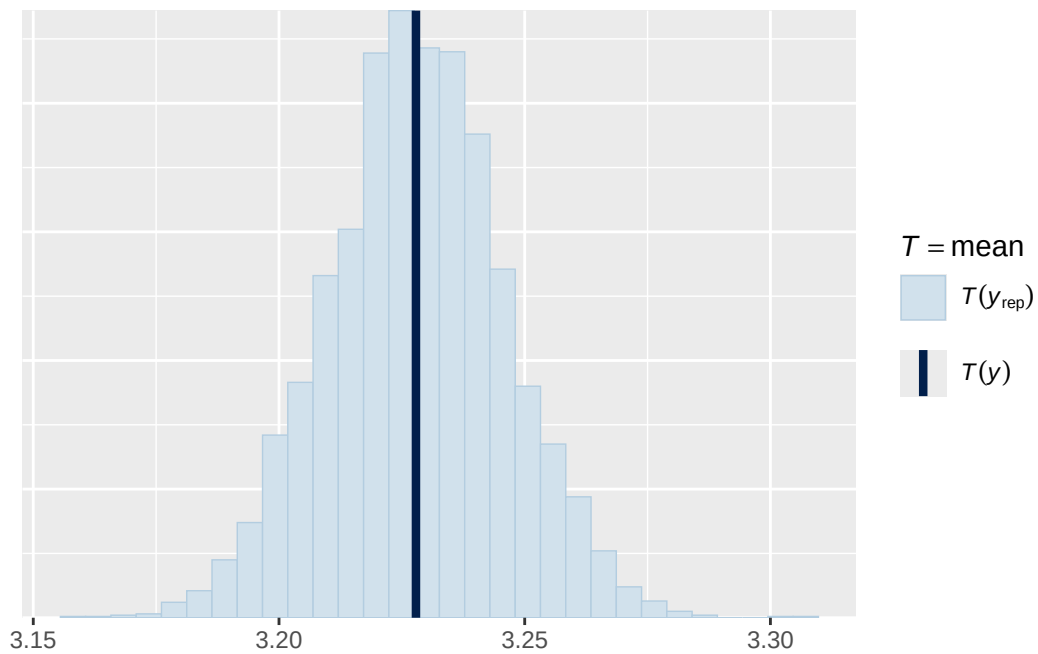
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



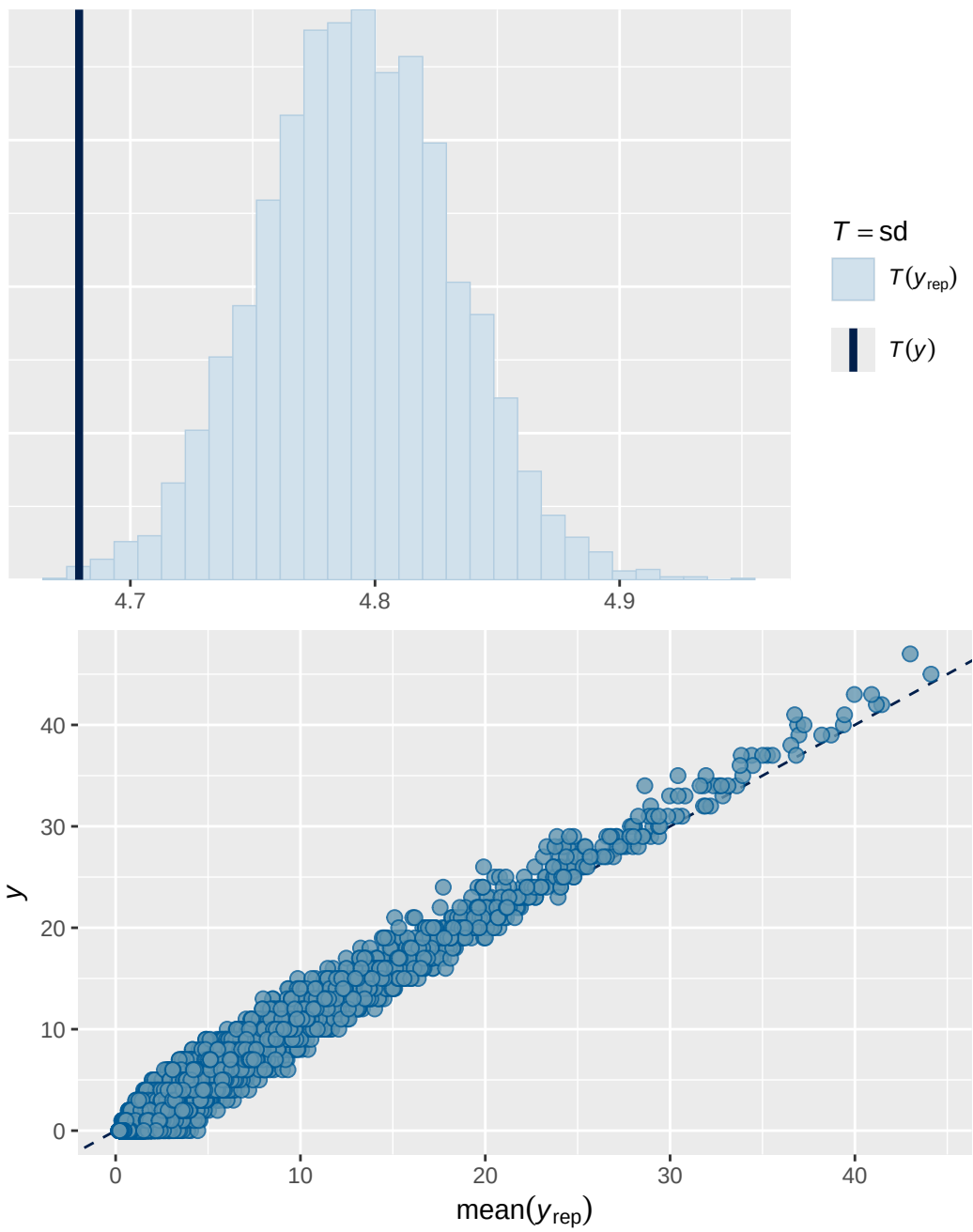
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 3b output

```

      Estimate Est.Error  Q2.5 Q97.5
R2      0.859      0.00256 0.854 0.864

```

```

Family: poisson
Links: mu = log
Formula: externalising ~ breast_development_p_sc * adhd_status_life + age_sc + ethnrace + inr_sc
Data: analytic_df (Number of observations: 19358)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 4613)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.859      0.031    0.798    0.917 1.001      822    1858

```

```

~obs_id (Number of levels: 19358)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.494      0.008    0.478    0.510 1.001     3006     5123

```

```

~participant_id (Number of levels: 5283)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.769      0.031    0.710    0.830 1.001      732     1812

```

```

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.153      0.034    0.097    0.231 1.000     5190     5332

```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI
Intercept	0.156	0.045	0.066	0.245
breast_development_p_sc	0.148	0.025	0.099	0.197
adhd_status_lifeY	1.320	0.044	1.235	1.405
age_sc	-0.149	0.019	-0.186	-
ethnraceblack_nh	-0.014	0.058	-0.125	0.102
ethnracehispanic	-0.005	0.054	-0.110	0.101
ethnraceother	-0.013	0.060	-0.129	0.104
inr_sc	-0.174	0.027	-0.226	-

breast_development_p_sc:adhd_status_lifeY	-0.064	0.037	-0.136	0.008
	Rhat	Bulk_ESS	Tail_ESS	
Intercept	1.000	4898	5183	
breast_development_p_sc	1.000	7220	6679	
adhd_status_lifeY	1.000	7073	5690	
age_sc	1.000	7978	6418	
ethnraceblack_nh	1.000	6482	6244	
ethnracehispanic	1.001	6871	5548	
ethnraceother	1.000	6827	5715	
inr_sc	1.000	8890	6277	
breast_development_p_sc:adhd_status_lifeY	1.000	7550	5964	

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 3b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

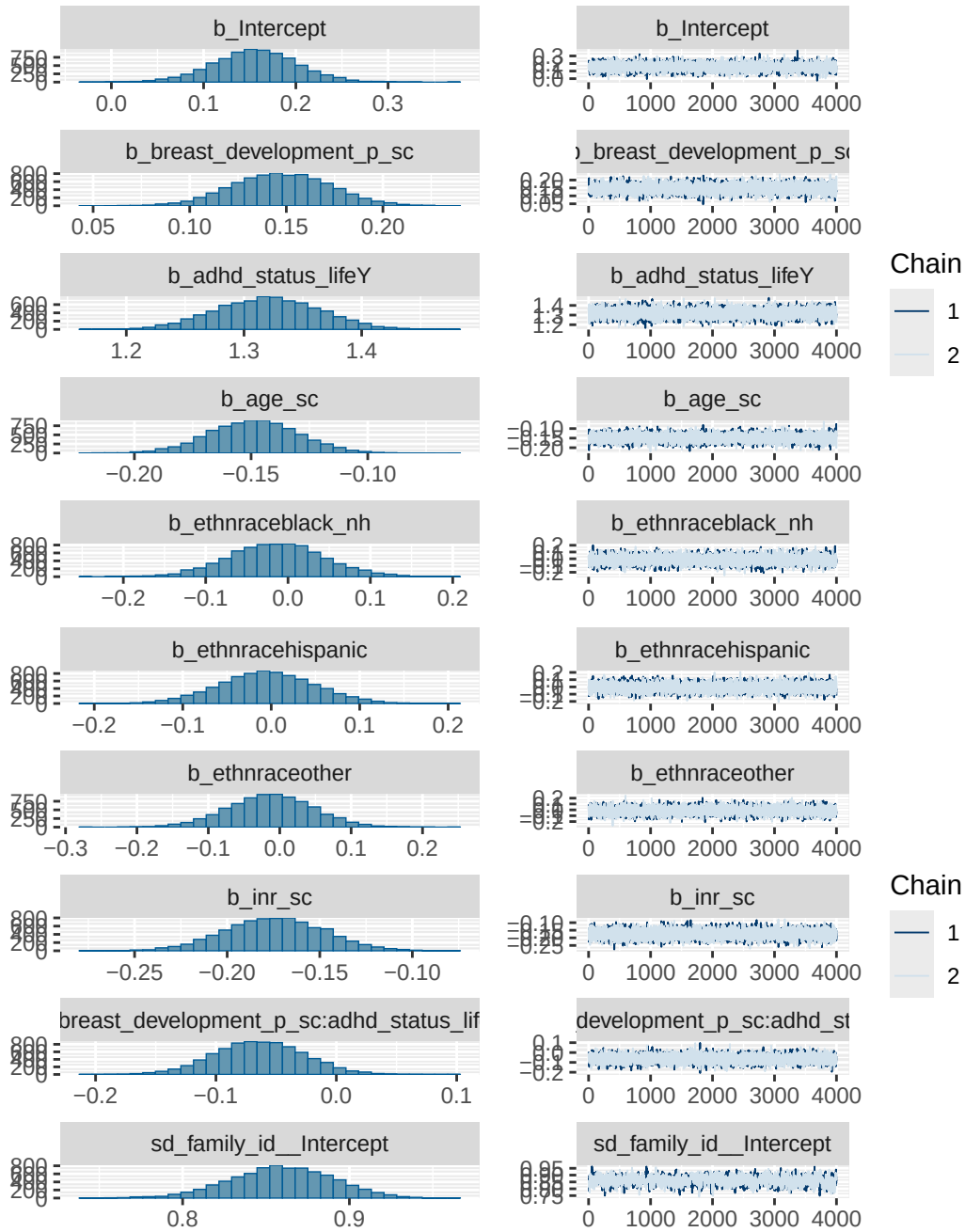
```
# A tibble: 3 x 4
```

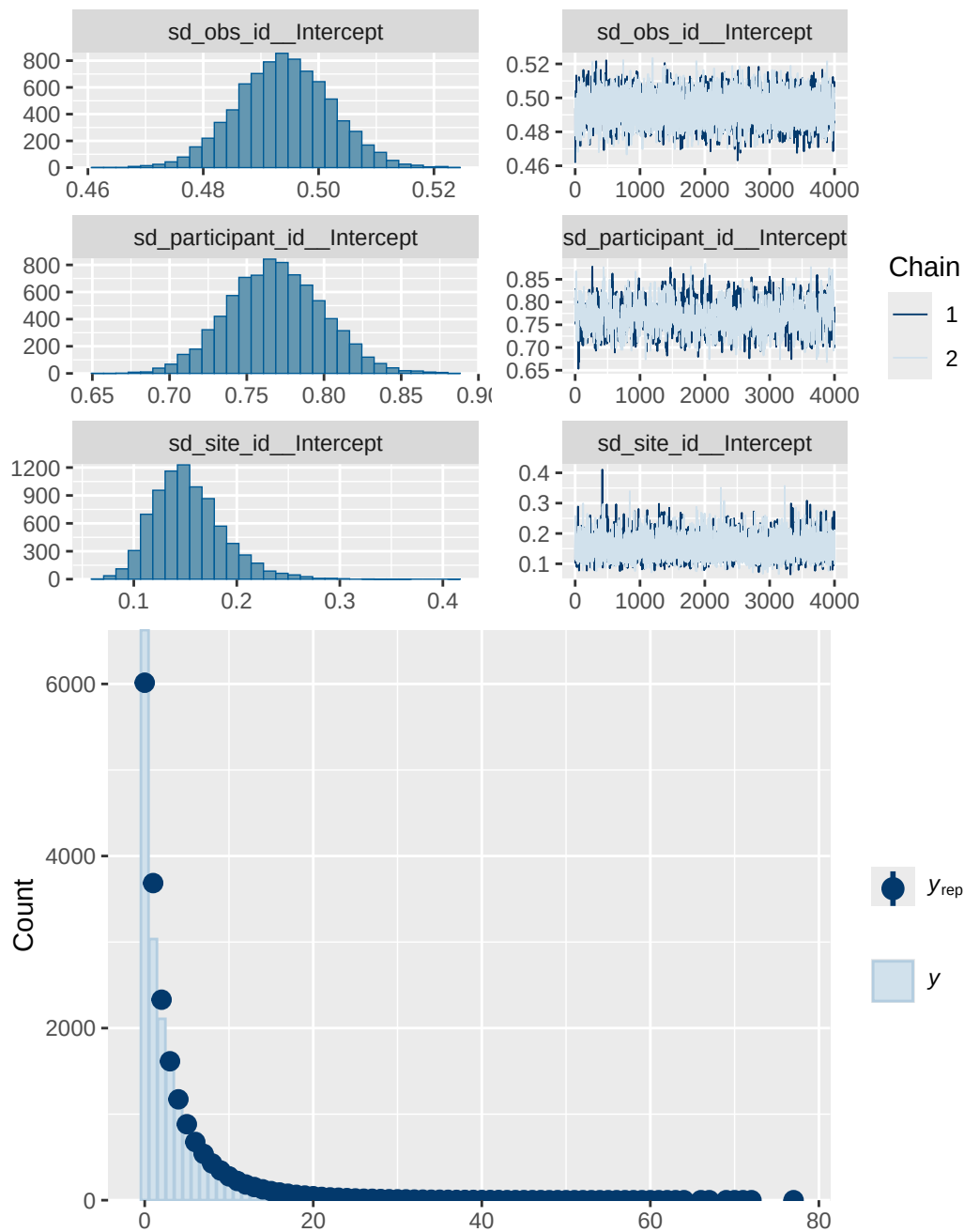
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development (per 1-unit) [ADHD ~	9.51	6.3	12.9
2 ADHD (yes vs no) [Breast development =~	274.	244.	307.
3 Interaction (Breast development × ADHD)	-3.86	-8.01	0.49

Model 3b diagnostics

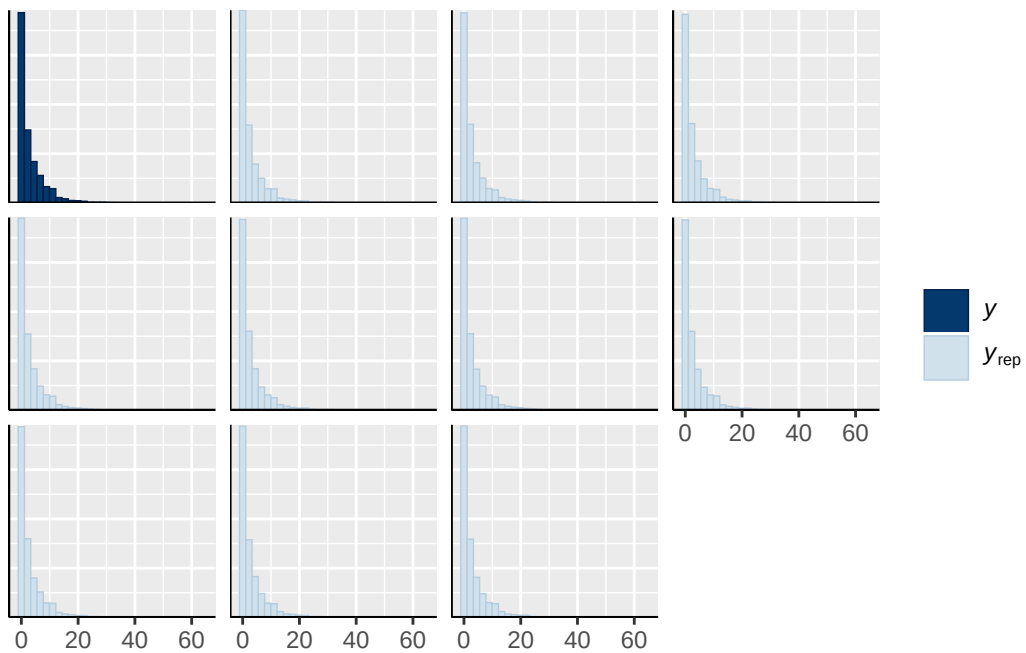
prior	class	coef
normal(0, 1)	b	
normal(0, 1)	b	adhd_status_lifeY
normal(0, 1)	b	age_sc
normal(0, 1)	b	breast_development_p_sc
normal(0, 1)	b	breast_development_p_sc:adhd_status_lifeY
normal(0, 1)	b	ethnraceblack_nh
normal(0, 1)	b	ethnracehispanic
normal(0, 1)	b	ethnraceother
normal(0, 1)	b	inr_sc
student_t(3, 0.7, 2.5)	Intercept	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	

student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
group resp dpar nlpar lb ub tag		source
		user
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
obs_id	0	(vectorized)
obs_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)

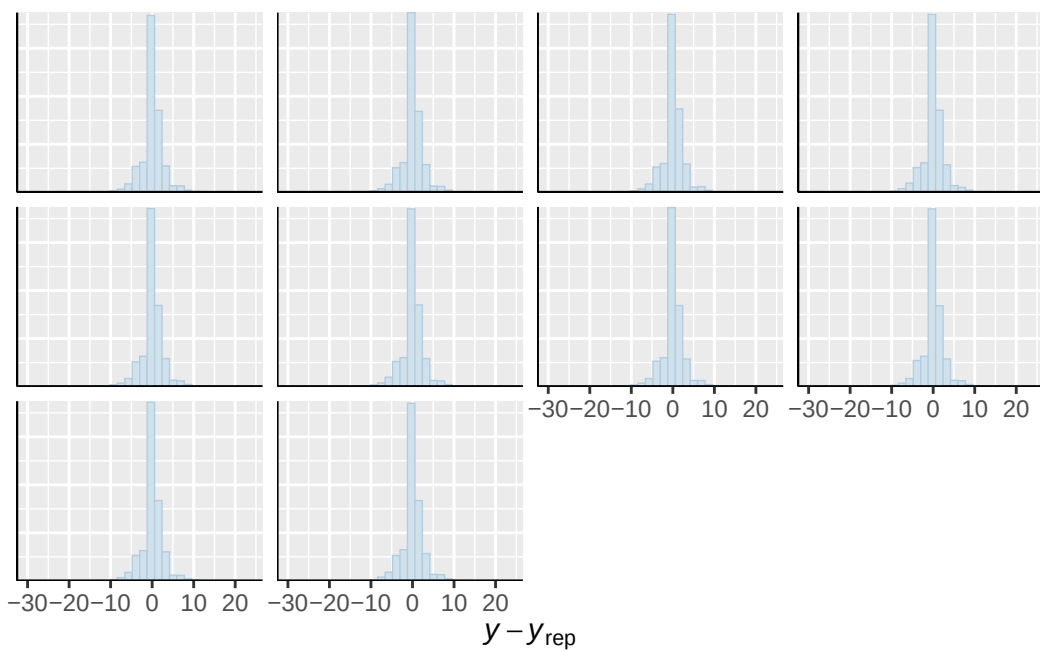




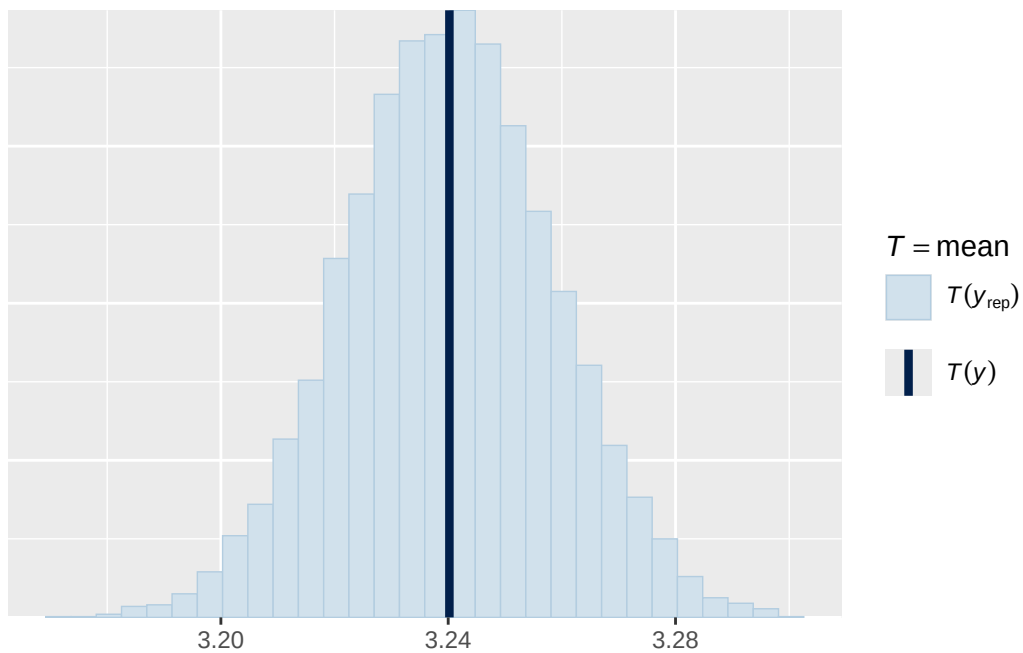
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



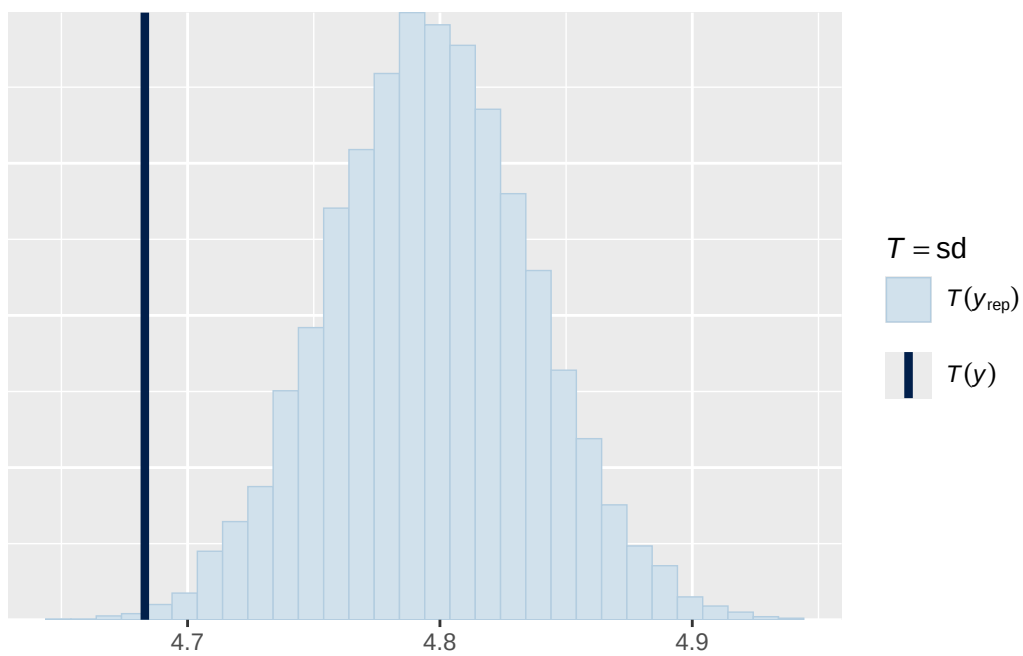
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

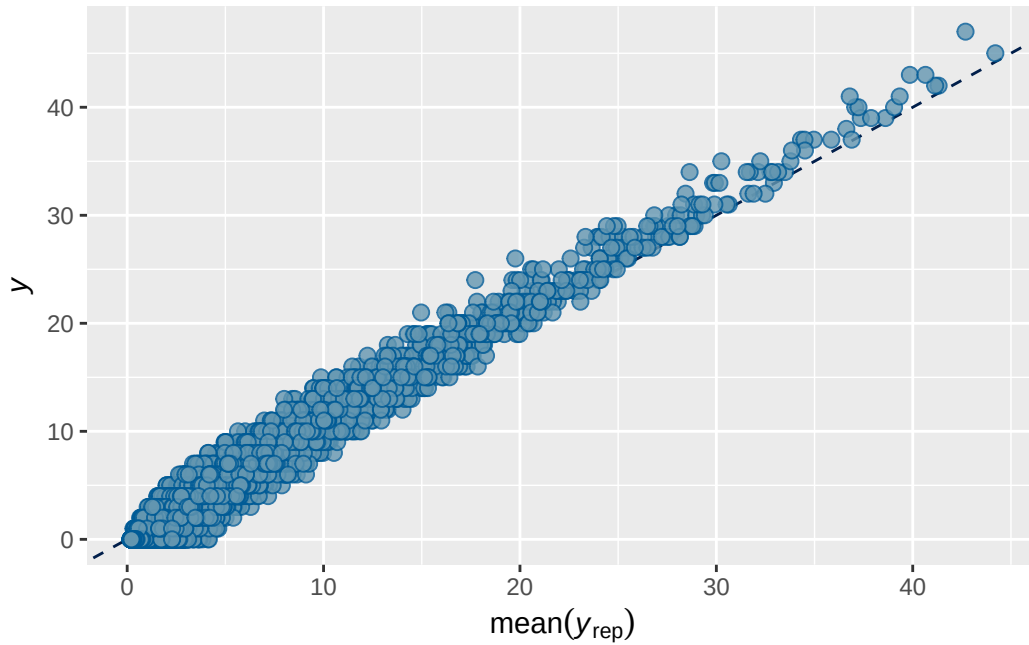


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Model 4 output

```

      Estimate Est.Error  Q2.5 Q97.5
R2      0.841    0.00237 0.836 0.845

```

Family: poisson

Links: mu = log

Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (

Data: analytic_df (Number of observations: 19381)

Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 4626)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.648	0.018	0.612	0.683	1.002	797	1729

~obs_id (Number of levels: 19381)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.367	0.006	0.354	0.379	1.000	2687	4390

~participant_id (Number of levels: 5297)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.462	0.020	0.423	0.501	1.005	585	1330

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.092	0.021	0.056	0.139	1.000	2747	4282

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
Intercept	1.200	0.028	1.144	1.254	1.000
menarche_status_pY	0.071	0.016	0.039	0.103	1.001
adhd_traits_sc	0.815	0.016	0.785	0.846	1.001
age_sc	0.057	0.014	0.030	0.084	1.000
ethnraceblack_nh	-0.438	0.040	-0.515	-0.360	1.001
ethnracehispanic	-0.080	0.037	-0.152	-0.007	1.000
ethnraceother	-0.046	0.042	-0.130	0.034	1.001
inr_sc	-0.023	0.018	-0.059	0.013	1.000
menarche_status_pY:adhd_traits_sc	0.041	0.020	0.001	0.081	1.000

	Bulk_ESS	Tail_ESS
Intercept	3147	3872
menarche_status_pY	5428	5483
adhd_traits_sc	4505	4806
age_sc	5473	5582
ethnraceblack_nh	4531	5349
ethnracehispanic	3486	4330
ethnraceother	4434	4942
inr_sc	5158	5553
menarche_status_pY:adhd_traits_sc	5025	5329

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 4 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

A tibble: 3 x 4

Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Menarche Y [ADHD traits = mean]	7.38	4.02	10.9
2 ADHD traits [Menarche = 0]	15.6	14.9	16.2
3 Menarche × ADHD traits	0.74	0.02	1.45

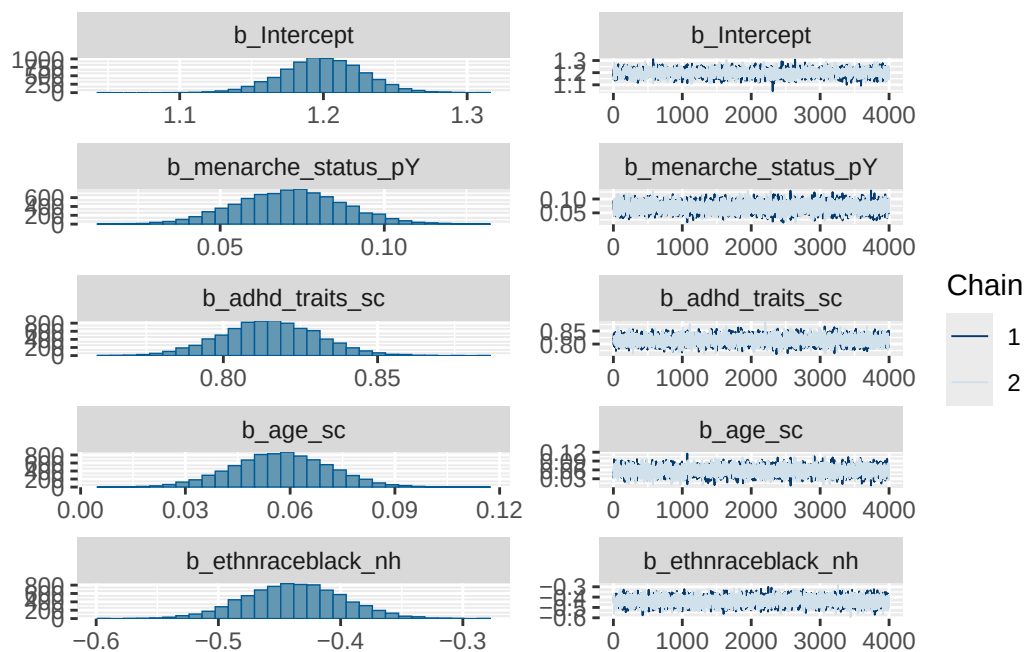
Model 4 interaction analysis

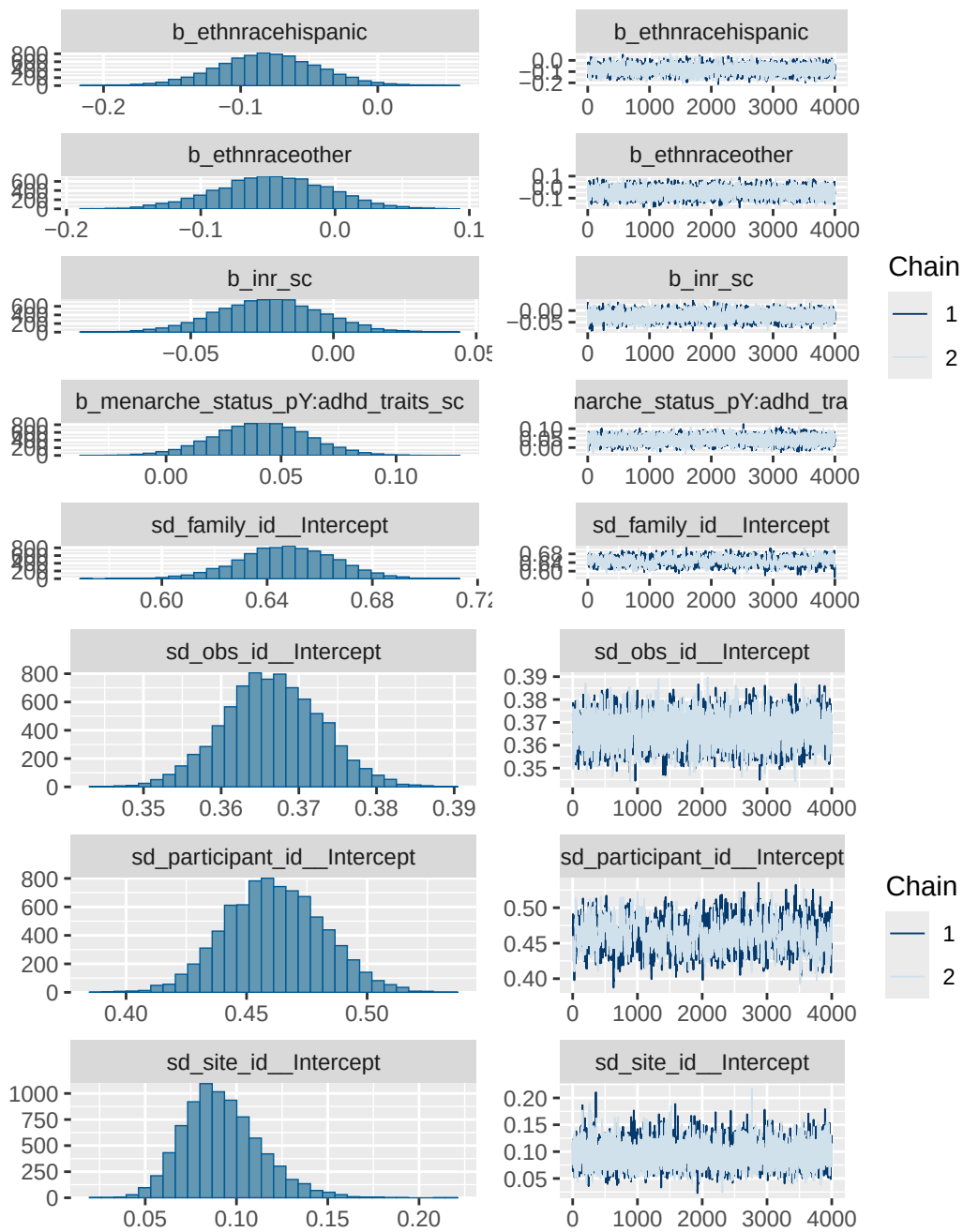
	menarche_status_p	Estimate	Lower 95% CI	Upper 95% CI
1	N	15.56	14.91	16.16
2	Y	16.41	15.64	17.24

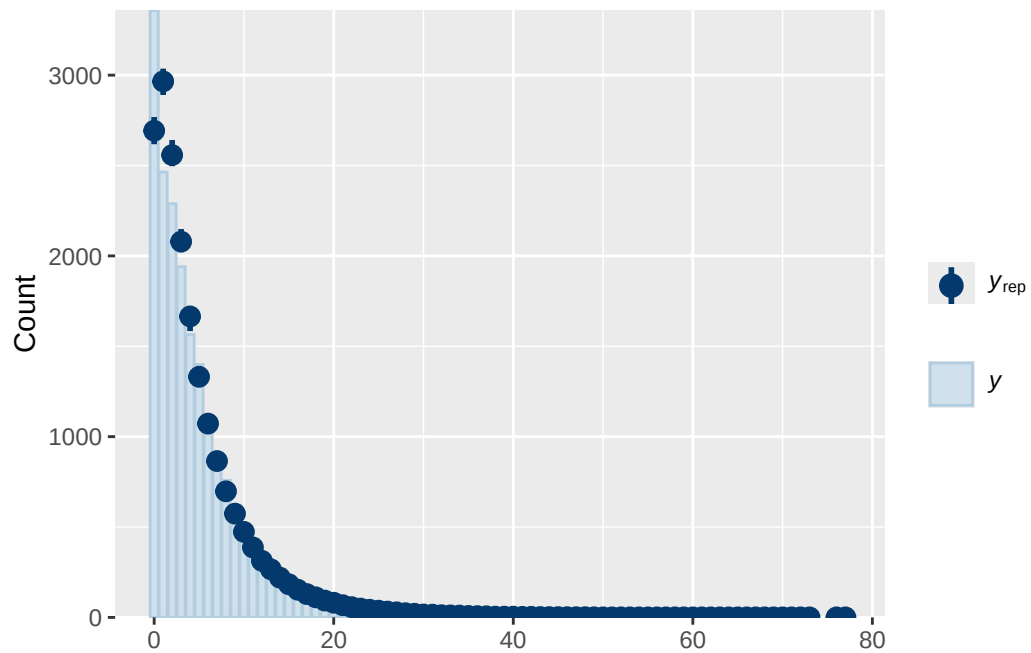
Model 4 diagnostics

	prior	class	coef
	normal(0, 1)	b	
	normal(0, 1)	b	adhd_traits_sc
	normal(0, 1)	b	age_sc
	normal(0, 1)	b	ethnraceblack_nh
	normal(0, 1)	b	ethnracehispanic
	normal(0, 1)	b	ethnraceother
	normal(0, 1)	b	inr_sc
	normal(0, 1)	b	menarche_status_pY
	normal(0, 1)	b	menarche_status_pY:adhd_traits_sc
student_t(3, 1.1, 2.5)	Intercept		
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		Intercept
group resp dpar nlpar lb ub tag			source
			user
			(vectorized)
			(vectorized)
			(vectorized)
			(vectorized)
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family_id		0	(vectorized)
family_id		0	(vectorized)

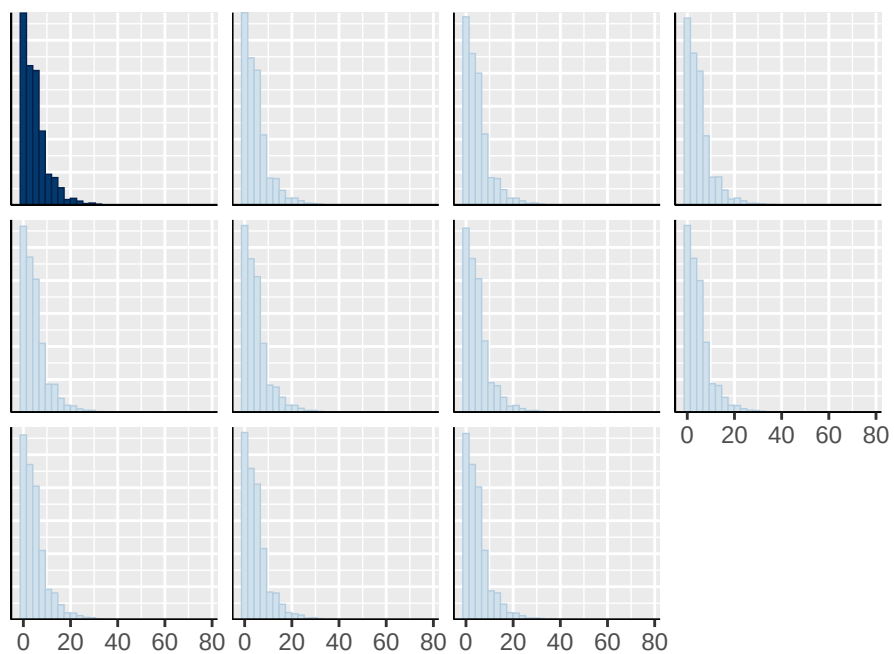
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obs_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



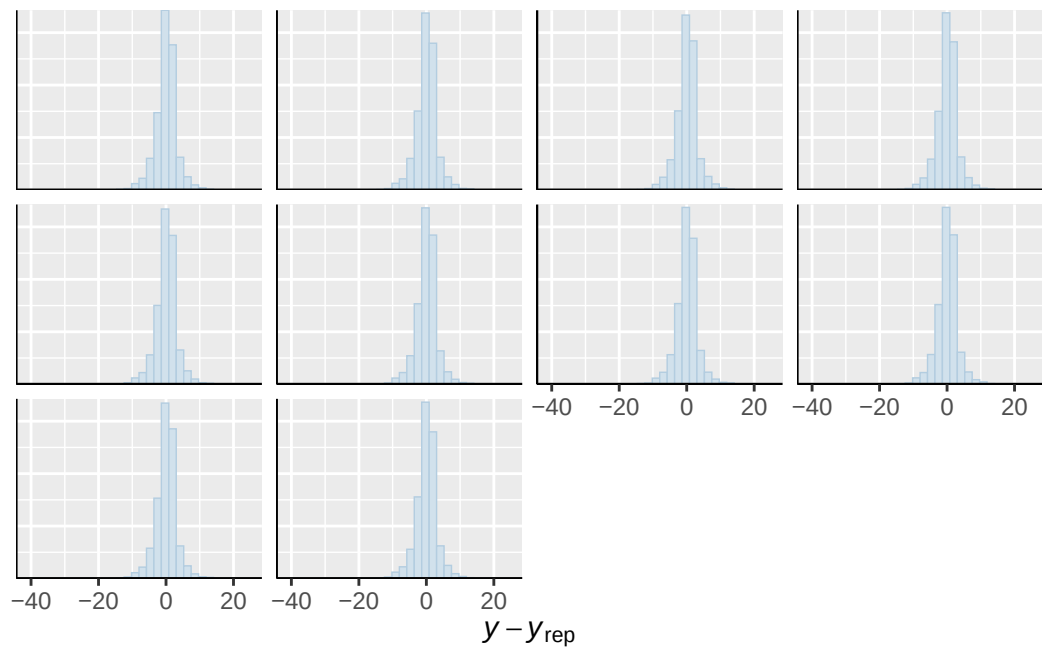




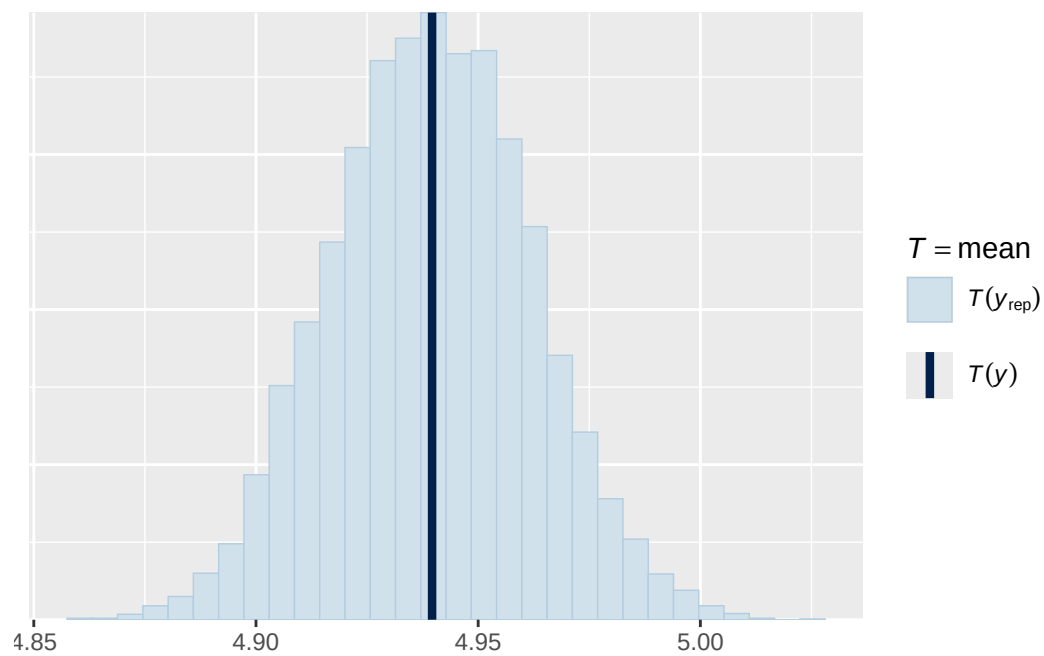
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



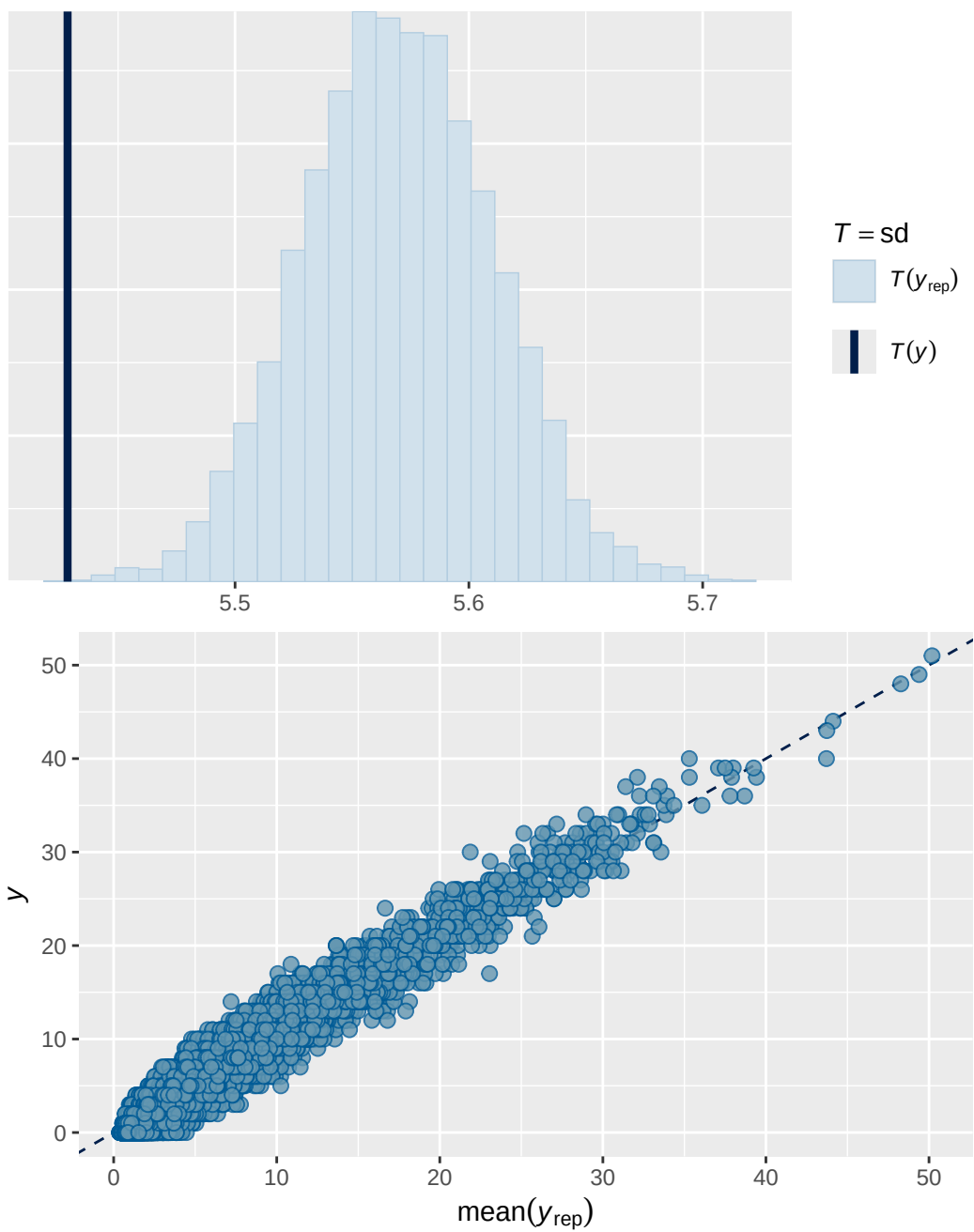
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 4b output

Estimate Est.Error Q2.5 Q97.5
R2 0.841 0.0024 0.836 0.845

Family: poisson
Links: mu = log
Formula: internalising ~ breast_development_p_sc * adhd_traits_sc + age_sc + ethnrace + inr_sc
Data: analytic_df (Number of observations: 19364)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 4619)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.649	0.018	0.614	0.683	1.000	1116	2247

~obs_id (Number of levels: 19364)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.366	0.006	0.354	0.378	1.000	3108	5114

~participant_id (Number of levels: 5289)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.461	0.020	0.423	0.500	1.000	850	1715

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.092	0.022	0.055	0.142	1.000	3971	4763

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	1.221	0.028	1.162	1.274
breast_development_p_sc	0.057	0.017	0.025	0.089
adhd_traits_sc	0.827	0.014	0.799	0.854
age_sc	0.069	0.014	0.042	0.096
ethnraceblack_nh	-0.431	0.040	-0.509	-0.353
ethnracehispanic	-0.074	0.036	-0.146	-0.003
ethnraceother	-0.046	0.040	-0.126	0.033
inr_sc	-0.019	0.019	-0.056	0.018
breast_development_p_sc:adhd_traits_sc	0.021	0.021	-0.020	0.063

	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.001	4055	4623

breast_development_p_sc	1.000	6862	6647
adhd_traits_sc	1.000	6662	5946
age_sc	1.000	6712	5885
ethnraceblack_nh	1.000	5460	5714
ethnracehispanic	1.001	4816	4652
ethnraceother	1.001	4712	5239
inr_sc	1.000	7281	5925
breast_development_p_sc:adhd_traits_sc	1.000	7014	6048

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 4b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

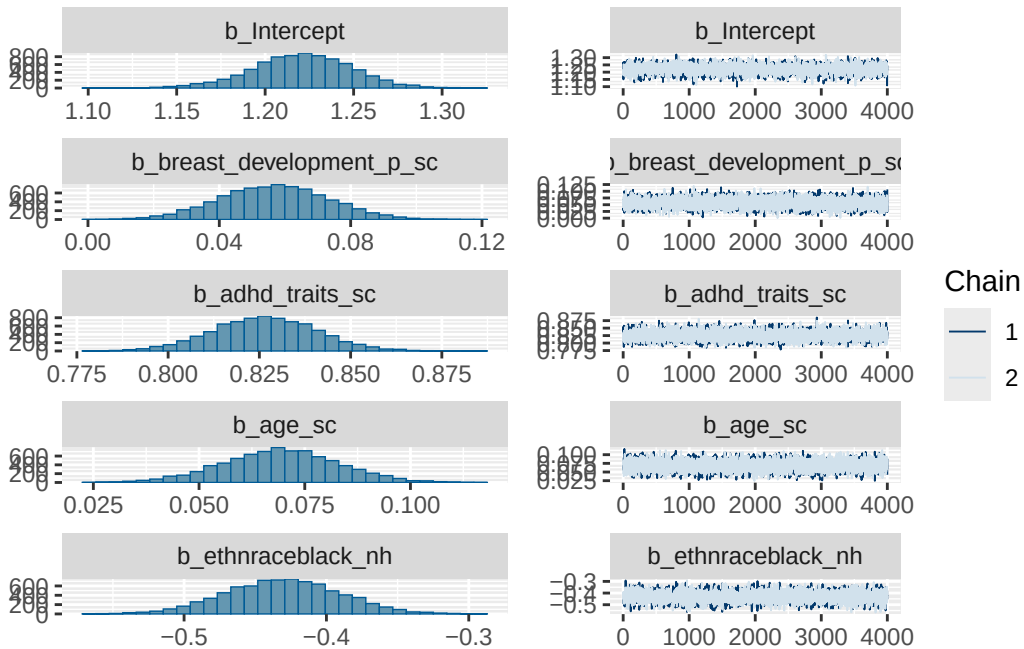
```
# A tibble: 3 x 4
```

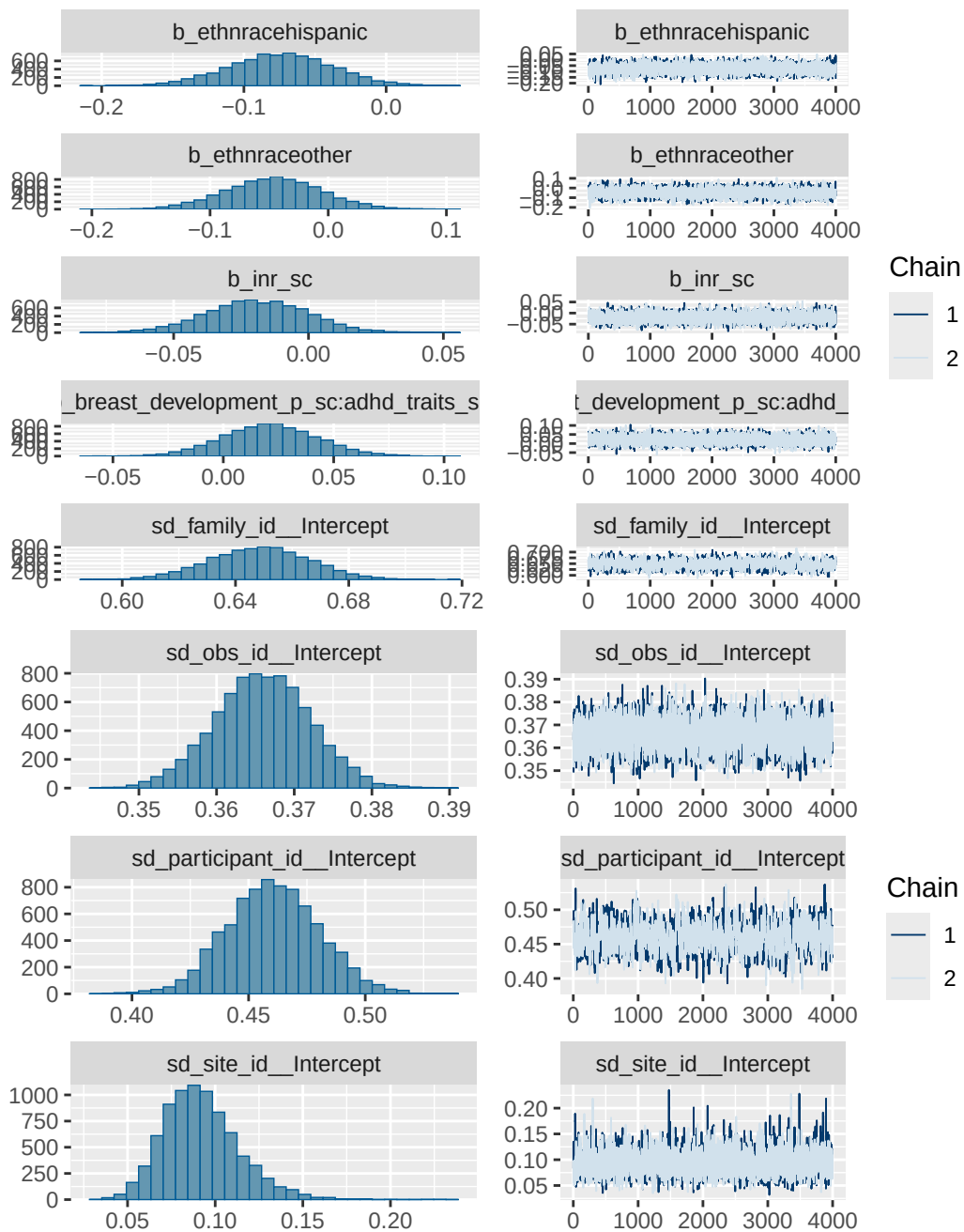
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development (per 1-unit) [ADHD ~	3.55	1.55	5.65
2 ADHD traits (per 1-unit) [Breast devel~	15.8	15.2	16.4
3 Interaction (Breast development × ADHD~	0.23	-0.22	0.69

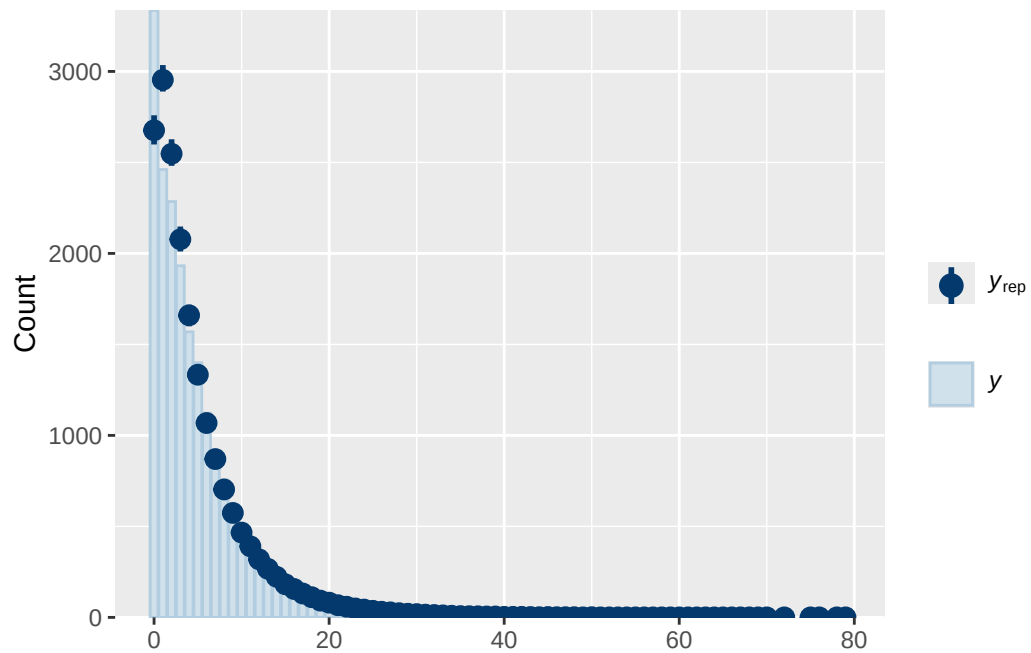
Model 4b diagnostics

prior	class	coef
normal(0, 1)	b	
normal(0, 1)	b	adhd_traits_sc
normal(0, 1)	b	age_sc
normal(0, 1)	b	breast_development_p_sc
normal(0, 1)	b	breast_development_p_sc:adhd_traits_sc
normal(0, 1)	b	ethnraceblack_nh
normal(0, 1)	b	ethnracehispanic
normal(0, 1)	b	ethnraceother
normal(0, 1)	b	inr_sc
student_t(3, 1.1, 2.5)	Intercept	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept

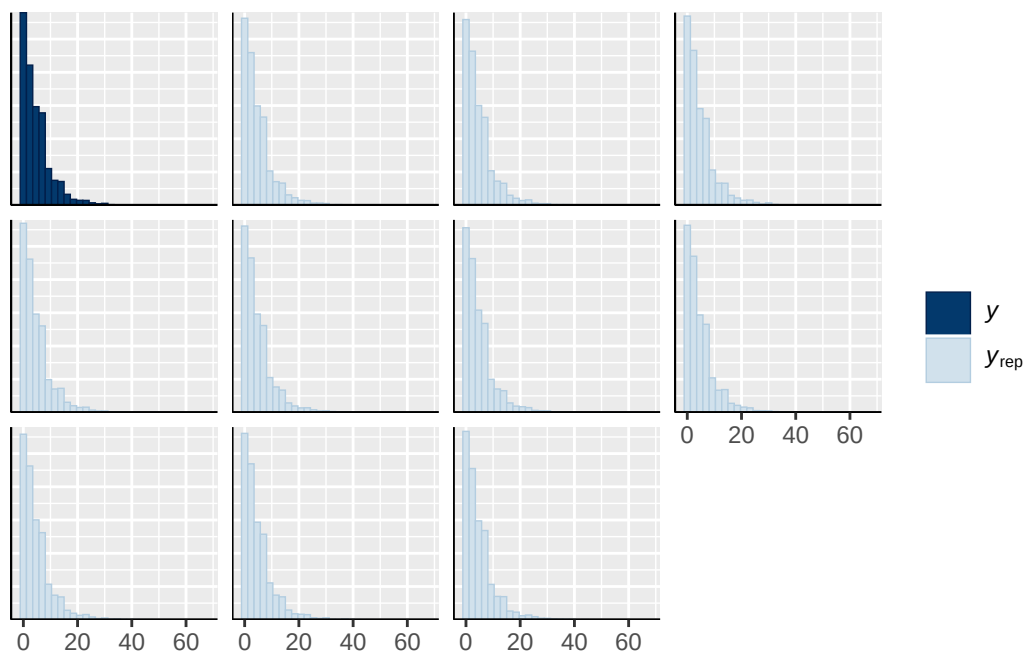
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
group resp dpar nlpar lb ub tag		source
		user
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
obs_id	0	(vectorized)
obs_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



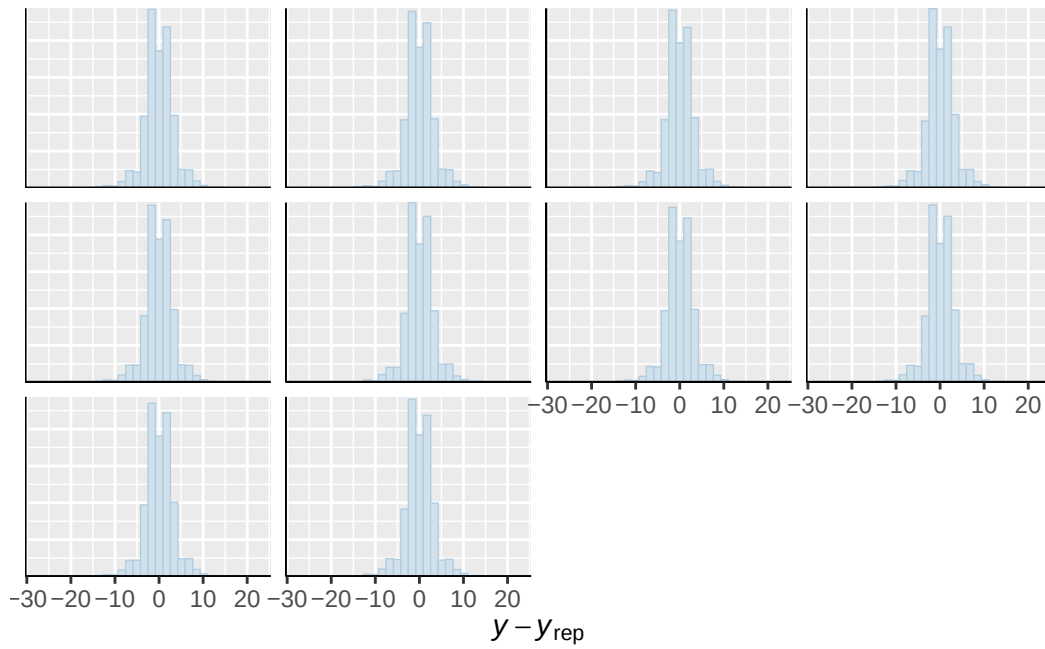




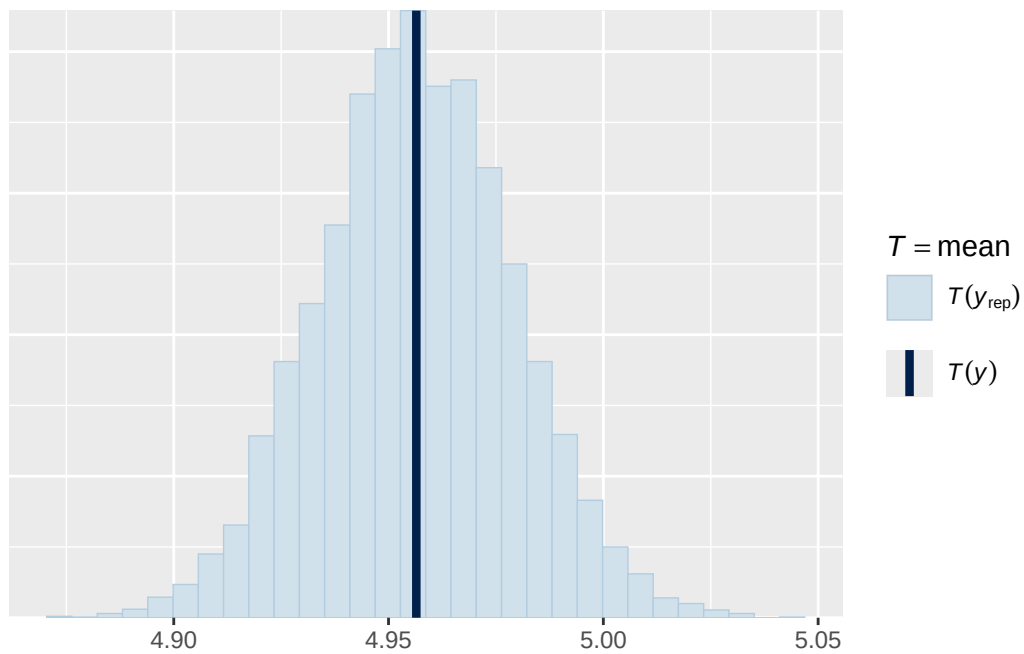
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



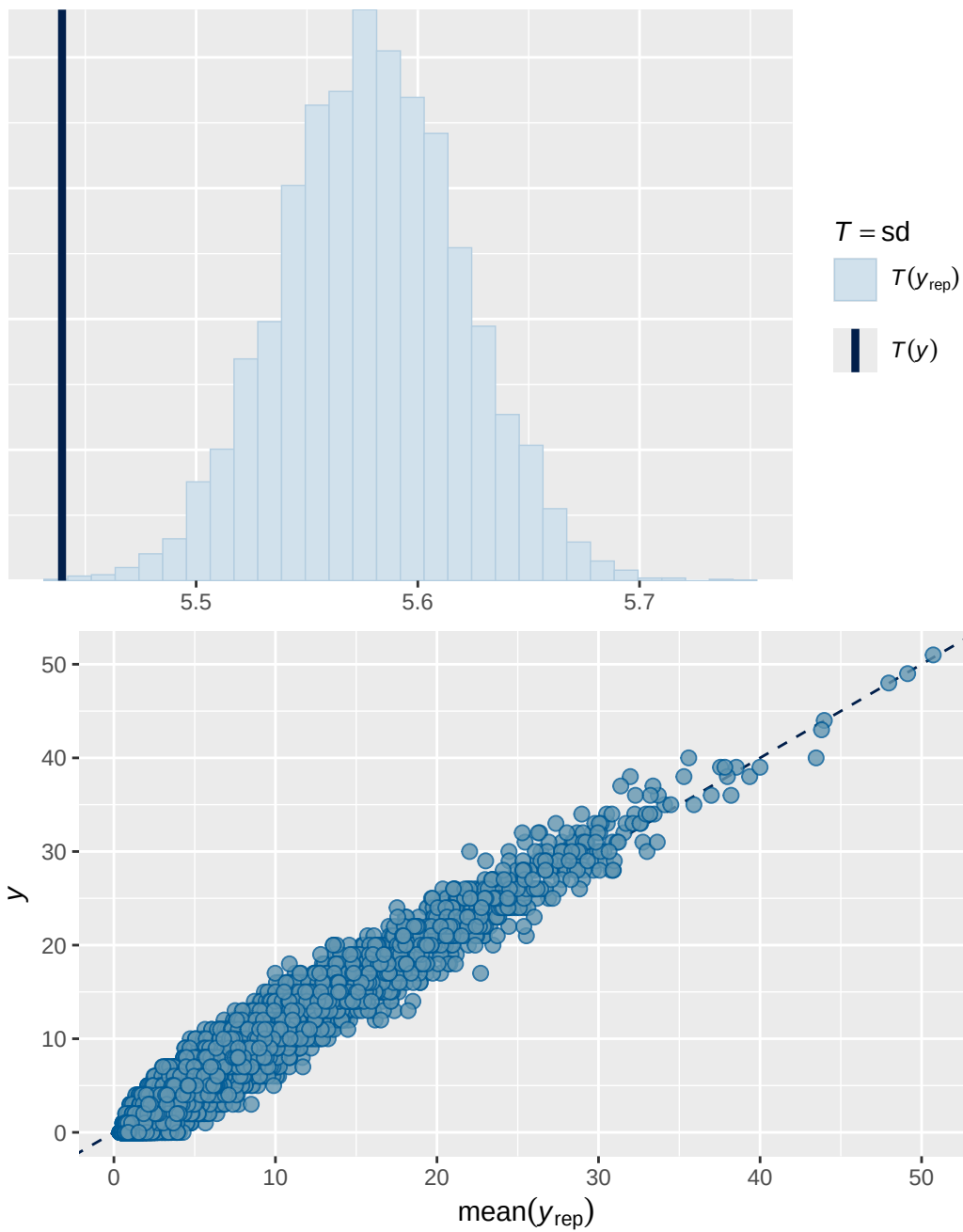
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 5 output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.863	0.00233	0.858	0.867

Family: poisson
 Links: mu = log
 Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (
 Data: analytic_df (Number of observations: 19381)
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 4626)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.719	0.031	0.656	0.776	1.013	465	919

~obs_id (Number of levels: 19381)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.373	0.008	0.357	0.389	1.000	1721	2867

~participant_id (Number of levels: 5297)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.724	0.028	0.673	0.781	1.013	486	826

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.111	0.027	0.065	0.172	1.000	3767	3204

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
Intercept	0.412	0.035	0.341	0.480	1.001
menarche_status_pY	0.079	0.020	0.039	0.118	1.001
adhd_traits_sc	1.077	0.019	1.040	1.114	1.000
age_sc	-0.123	0.017	-0.157	-0.089	1.000
ethnraceblack_nh	-0.067	0.052	-0.170	0.037	1.001
ethnracehispanic	-0.044	0.048	-0.138	0.048	1.001
ethnraceother	-0.052	0.053	-0.155	0.050	1.001
inr_sc	-0.152	0.024	-0.197	-0.103	1.000
menarche_status_pY:adhd_traits_sc	0.048	0.023	0.003	0.095	1.001

	Bulk_ESS	Tail_ESS
Intercept	4048	2842
menarche_status_pY	5891	3346
adhd_traits_sc	5134	3151
age_sc	6012	3444
ethnraceblack_nh	5231	2990
ethnracehispanic	4835	3020
ethnraceother	5475	3266

inr_sc	5788	2679
menarche_status_pY:adhd_traits_sc	5362	3285

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 5 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 3 x 4
  Variable                                Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                                <dbl>         <dbl>         <dbl>
1 Menarche Y [ADHD traits = mean]      8.17          3.95          12.6
2 ADHD traits [Menarche = 0]           21.1          20.3          21.9
3 Menarche × ADHD traits                0.86          0.06          1.7
```

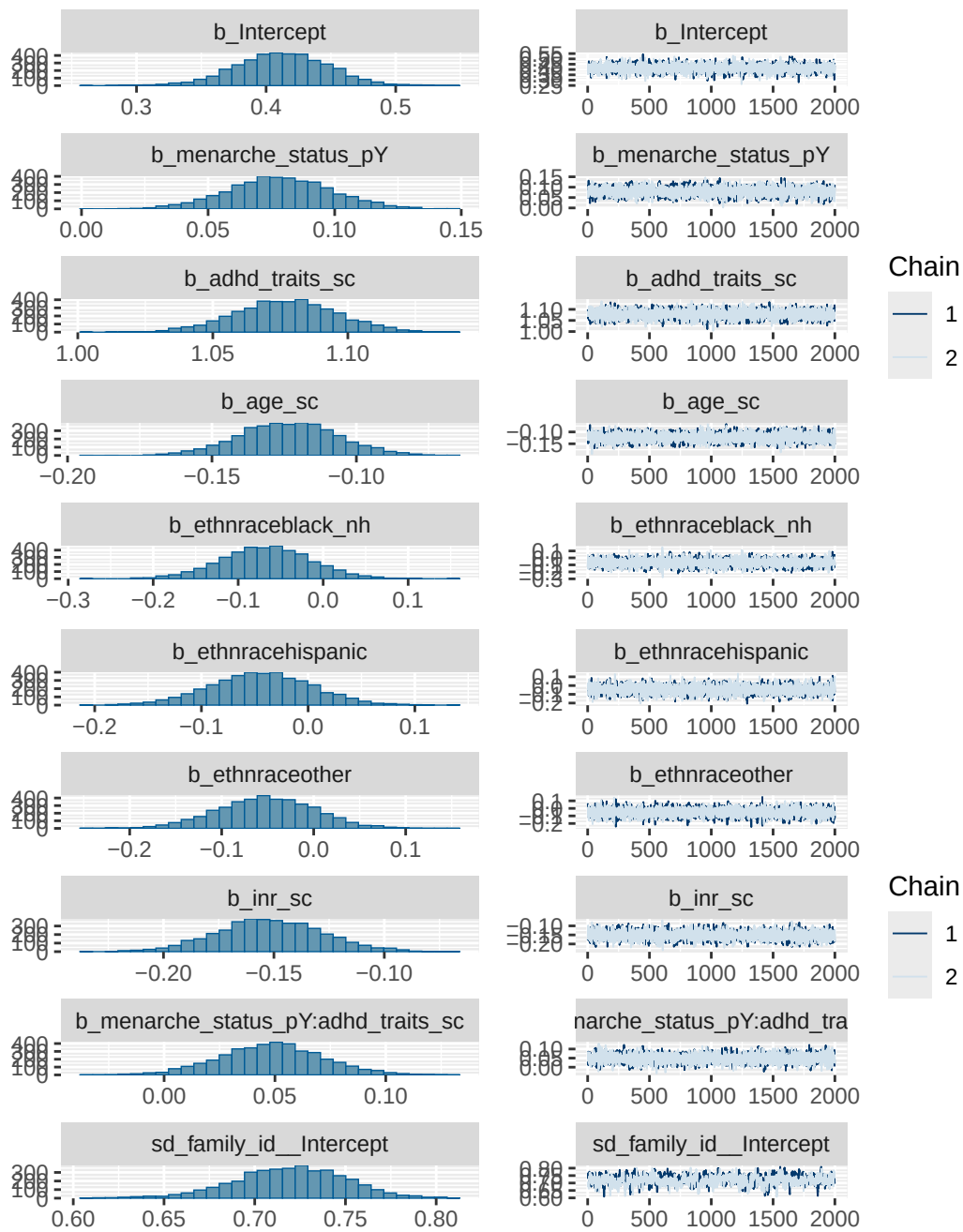
Model 5 interaction analysis

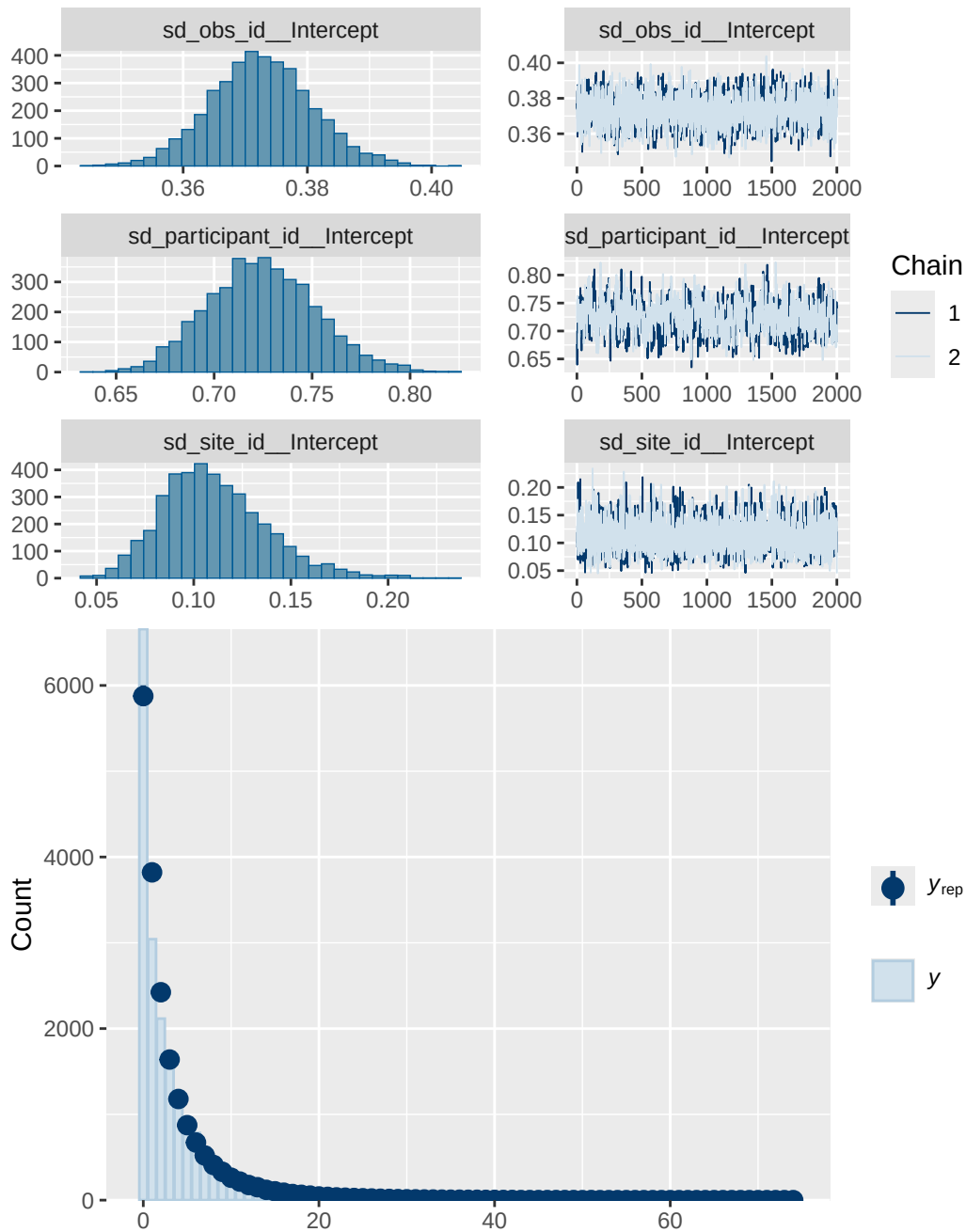
	menarche_status_p	Estimate	Lower 95% CI	Upper 95% CI
1	N	21.07	20.23	21.81
2	Y	22.12	21.17	23.17

Model 5 diagnostics

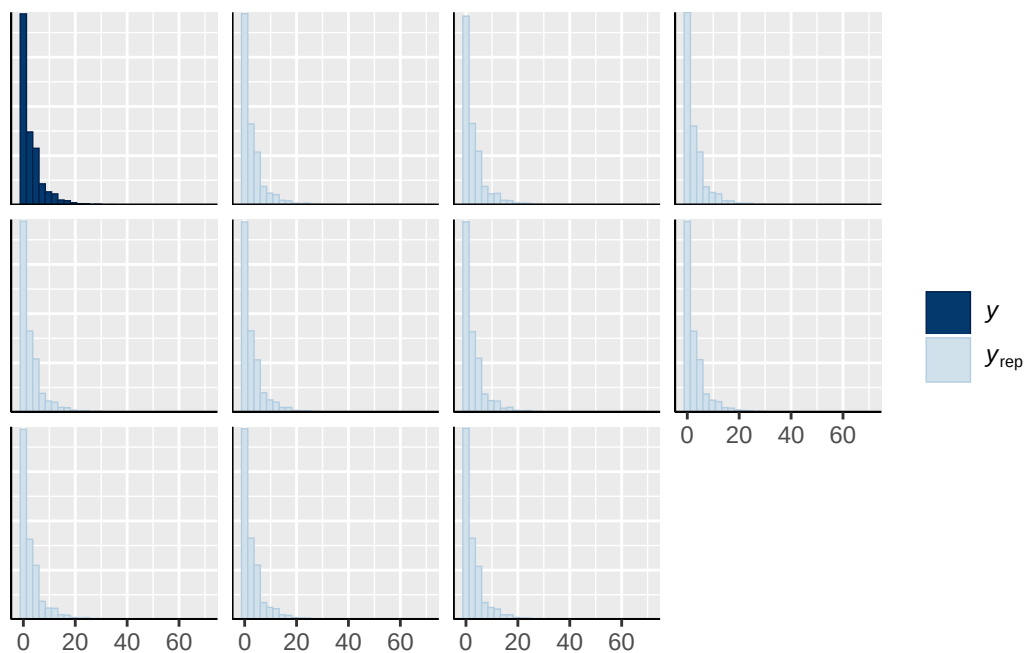
	prior	class	coef
	normal(0, 1)	b	
	normal(0, 1)	b	adhd_traits_sc
	normal(0, 1)	b	age_sc
	normal(0, 1)	b	ethnraceblack_nh
	normal(0, 1)	b	ethnracehispanic
	normal(0, 1)	b	ethnraceother
	normal(0, 1)	b	inr_sc
	normal(0, 1)	b	menarche_status_pY
	normal(0, 1)	b	menarche_status_pY:adhd_traits_sc
student_t(3, 0, 2.9)	Intercept		
student_t(3, 0, 2.9)		sd	
student_t(3, 0, 2.9)		sd	
student_t(3, 0, 2.9)		sd	Intercept
student_t(3, 0, 2.9)		sd	

student_t(3, 0, 2.9)	sd	Intercept
student_t(3, 0, 2.9)	sd	
student_t(3, 0, 2.9)	sd	Intercept
student_t(3, 0, 2.9)	sd	
student_t(3, 0, 2.9)	sd	Intercept
group resp dpar nlpar lb ub tag		source
		user
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
obs_id	0	(vectorized)
obs_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)

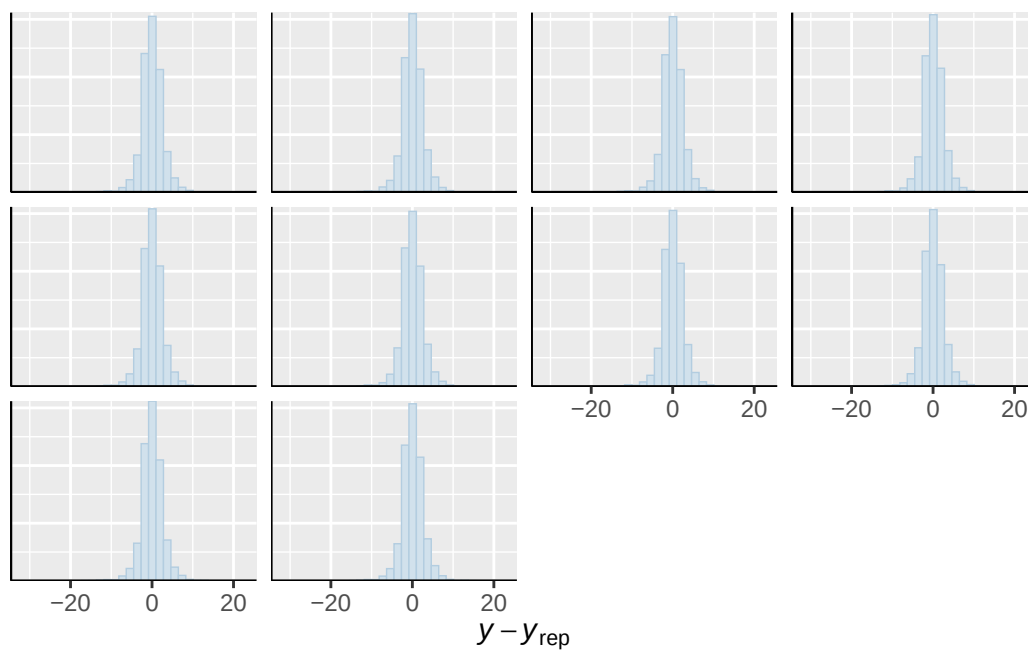




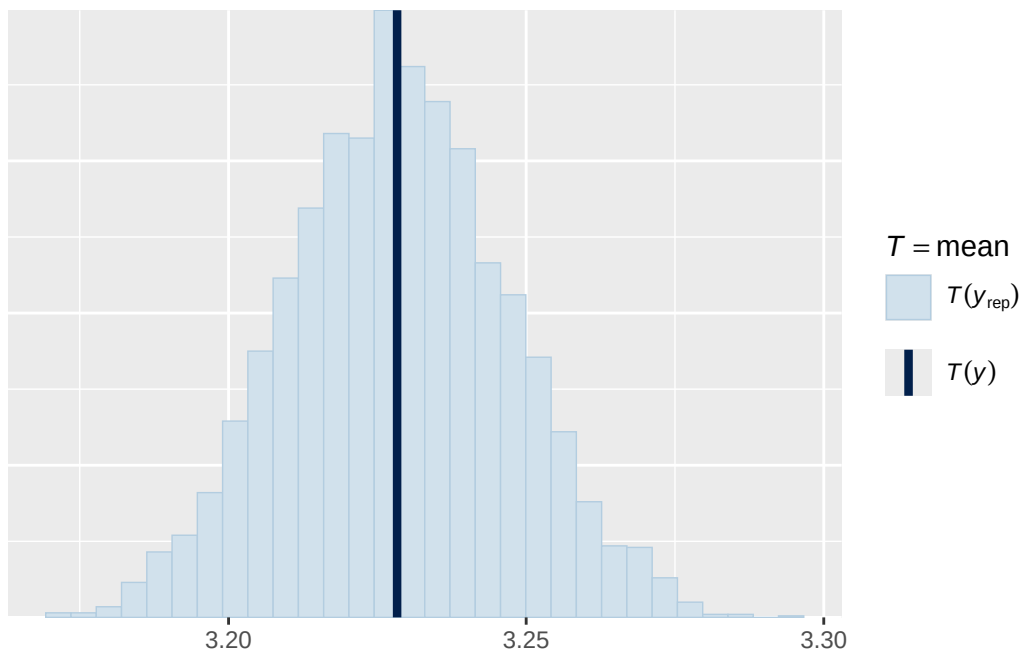
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



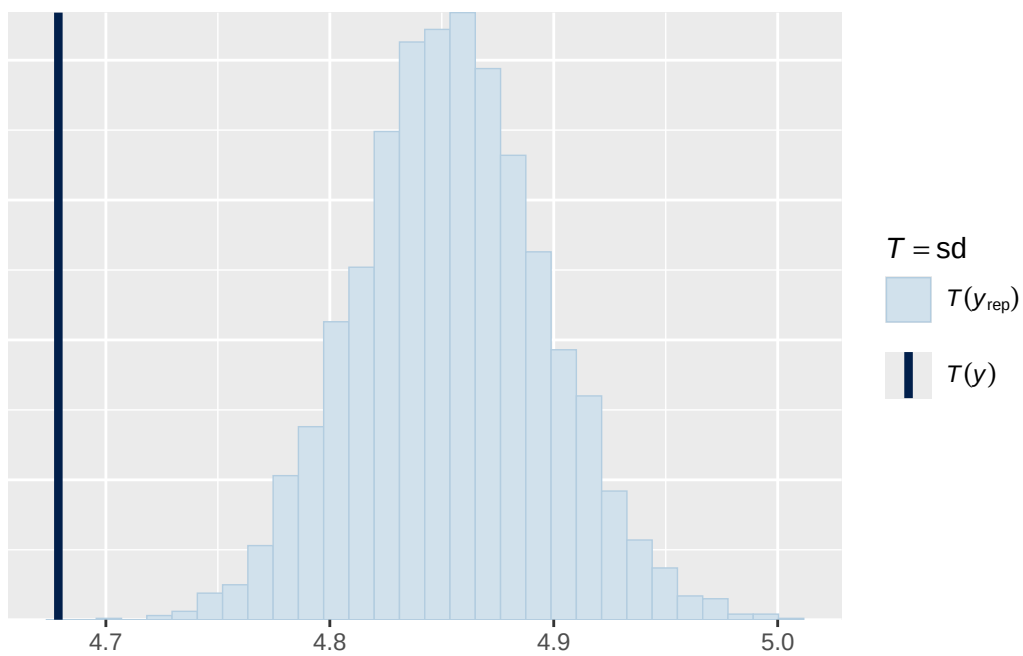
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

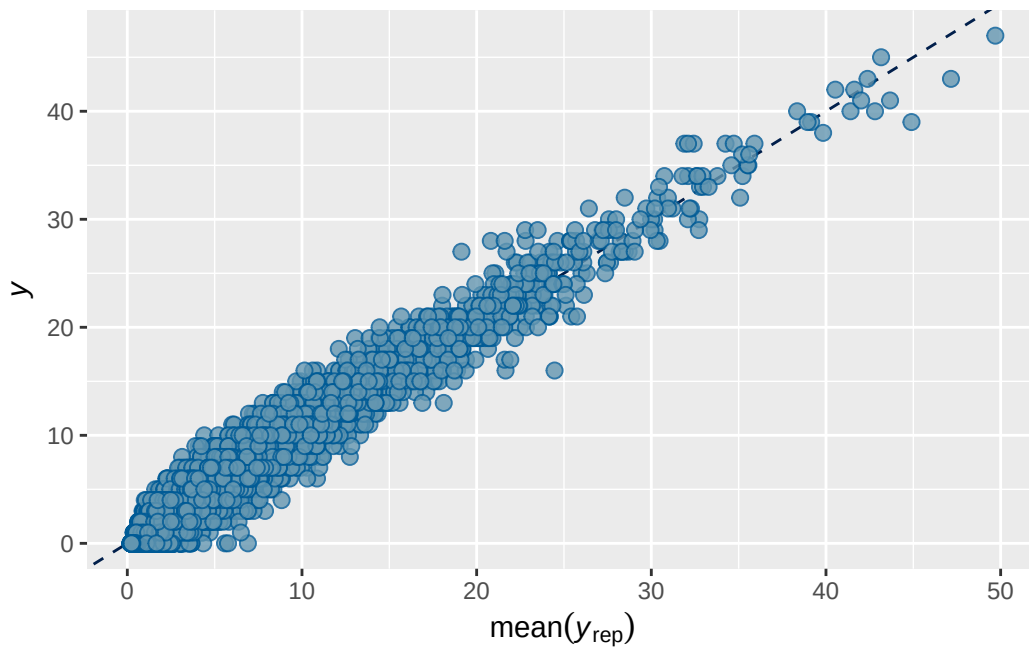


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Model 5b output

```

      Estimate Est.Error  Q2.5 Q97.5
R2    0.863    0.00233 0.858 0.867

```

```

Family: poisson
Links: mu = log

```

```

Formula: externalising ~ breast_development_p_sc * adhd_traits_sc + age_sc + ethnrace + inr_s

```

```

Data: analytic_df (Number of observations: 19364)

```

```

Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 4619)

```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.714	0.029	0.653	0.767	1.001	631	1017

```

~obs_id (Number of levels: 19364)

```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.369	0.008	0.353	0.385	1.002	1510	2684

```

~participant_id (Number of levels: 5289)

```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.728	0.028	0.676	0.784	1.002	649	866

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.112	0.027	0.066	0.172	1.001	3311	2815

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	0.441	0.036	0.369	0.512
breast_development_p_sc	0.109	0.020	0.069	0.150
adhd_traits_sc	1.089	0.017	1.056	1.124
age_sc	-0.128	0.017	-0.160	-0.097
ethnraceblack_nh	-0.078	0.052	-0.178	0.025
ethnracehispanic	-0.045	0.048	-0.138	0.047
ethnraceother	-0.058	0.053	-0.161	0.044
inr_sc	-0.149	0.023	-0.194	-0.104
breast_development_p_sc:adhd_traits_sc	-0.025	0.024	-0.070	0.023

	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.000	4227	2777
breast_development_p_sc	1.001	6233	3262
adhd_traits_sc	1.000	5412	3079
age_sc	1.003	5752	3257
ethnraceblack_nh	1.001	5533	2933
ethnracehispanic	1.001	4499	3261
ethnraceother	1.000	5564	2852
inr_sc	1.000	7177	3194
breast_development_p_sc:adhd_traits_sc	1.001	6426	3452

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

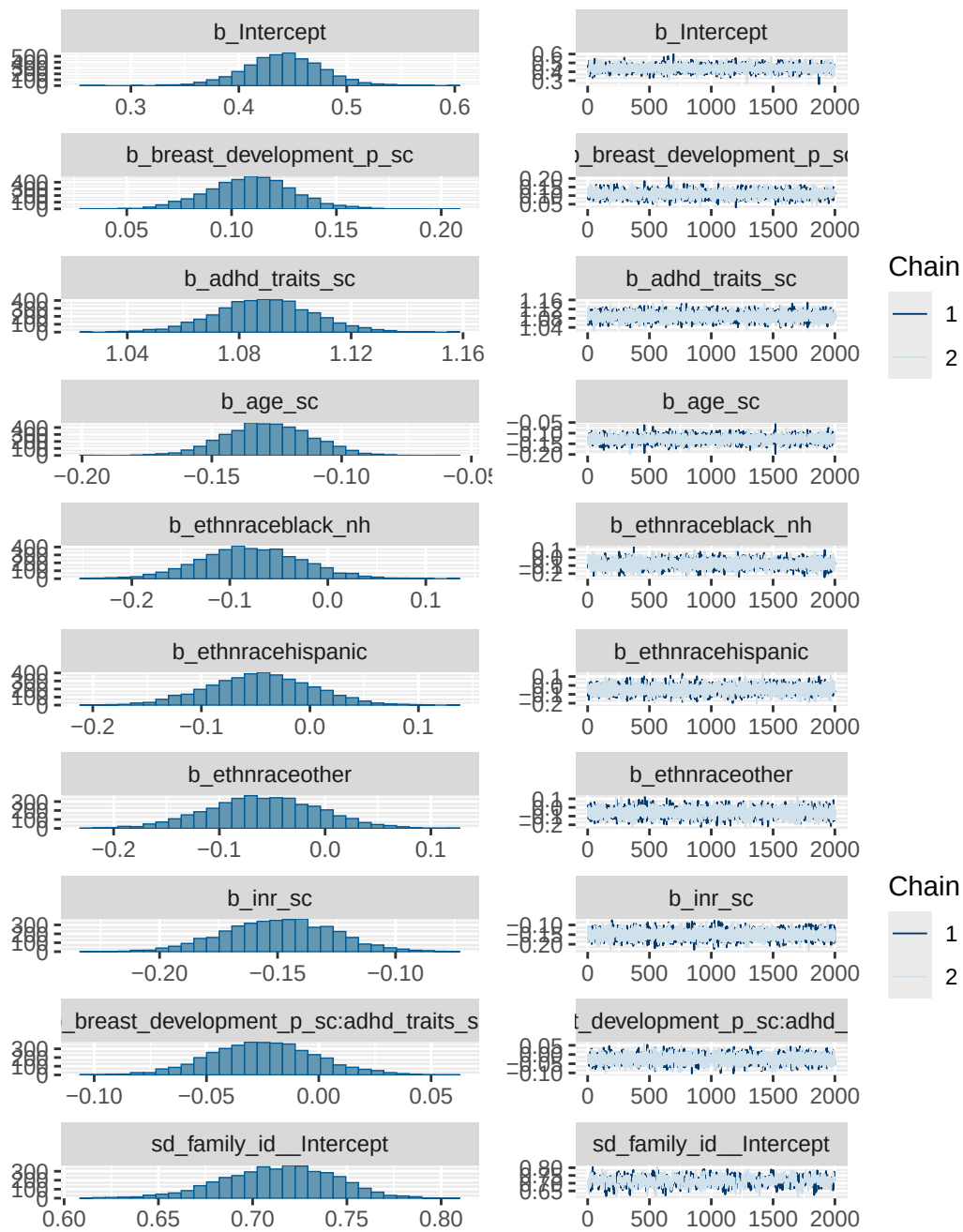
Model 5b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

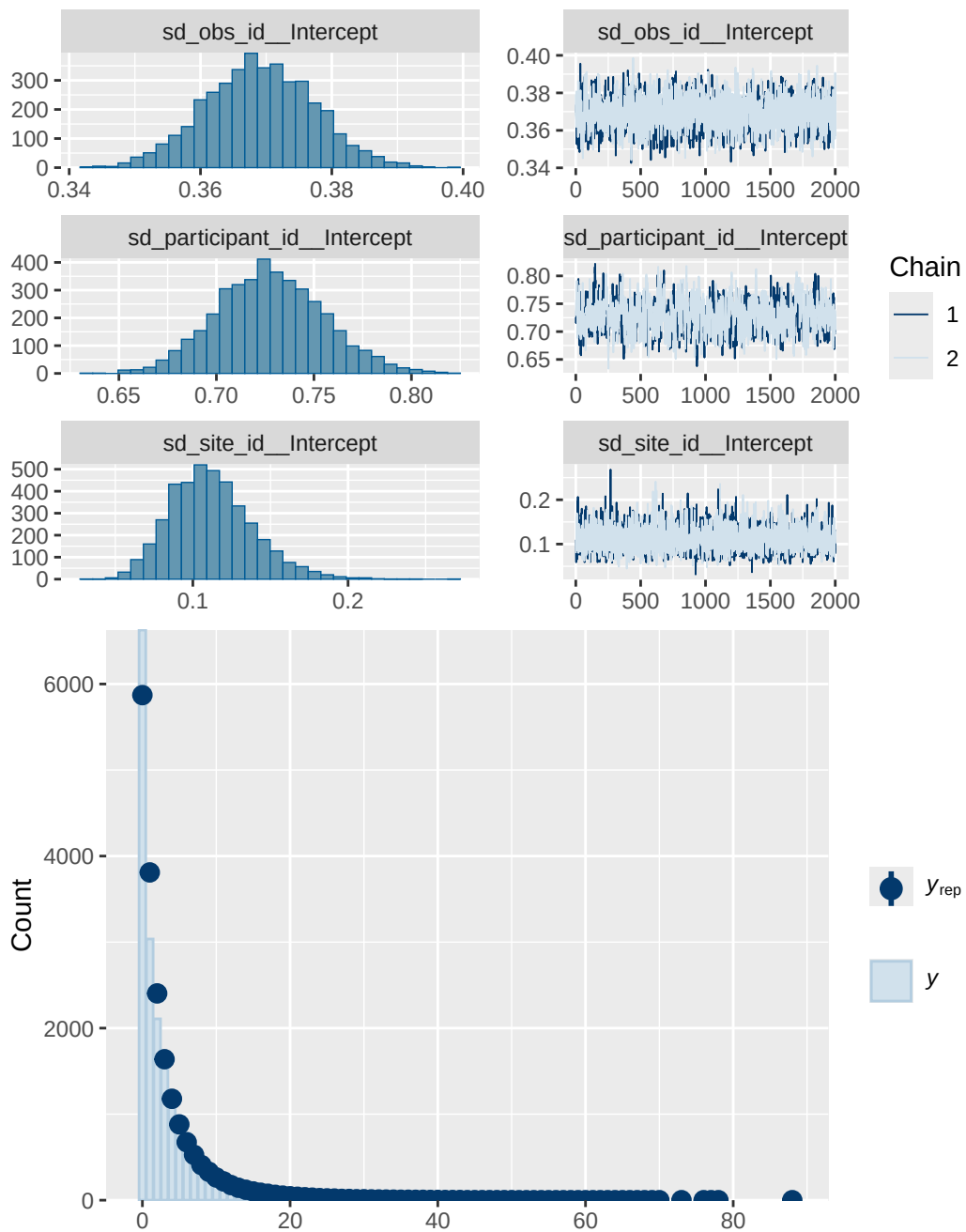
A tibble: 3 x 4

Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development (per 1-unit) [ADHD ~	6.96	4.34	9.69
2 ADHD traits (per 1-unit) [Breast devel~	21.3	20.6	22.1
3 Interaction (Breast development × ADHD~	-0.27	-0.76	0.25

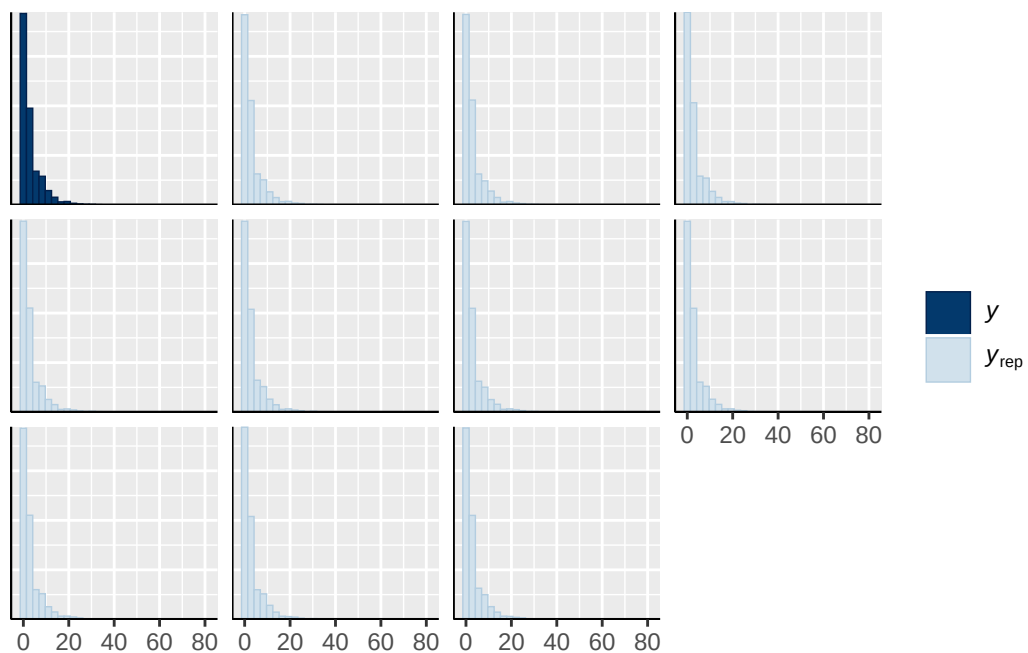
Model 5b diagnostics

	prior	class	coef
	normal(0, 1)	b	
	normal(0, 1)	b	adhd_traits_sc
	normal(0, 1)	b	age_sc
	normal(0, 1)	b	breast_development_p_sc
	normal(0, 1)	b	breast_development_p_sc:adhd_traits_sc
	normal(0, 1)	b	ethnraceblack_nh
	normal(0, 1)	b	ethnracehispanic
	normal(0, 1)	b	ethnraceother
	normal(0, 1)	b	inr_sc
student_t(3, 0.7, 2.5)	Intercept		
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		Intercept
group resp dpar nlpar lb ub tag			source
			user
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family_id		0	(vectorized)
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obs_id		0	(vectorized)
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participant_id		0	(vectorized)
site_id		0	(vectorized)
site_id		0	(vectorized)

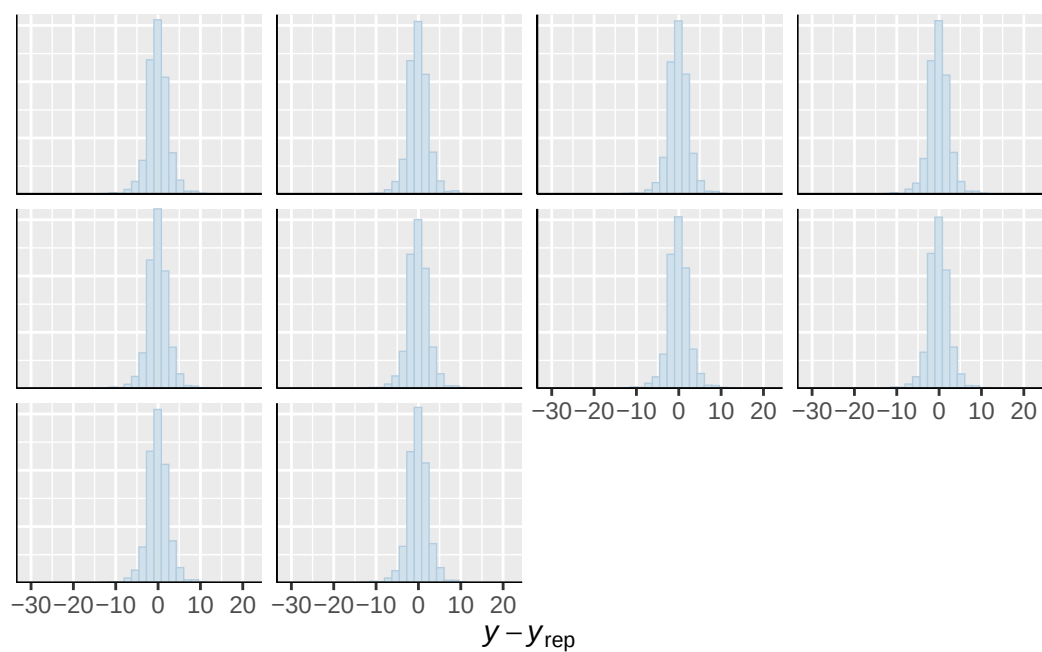




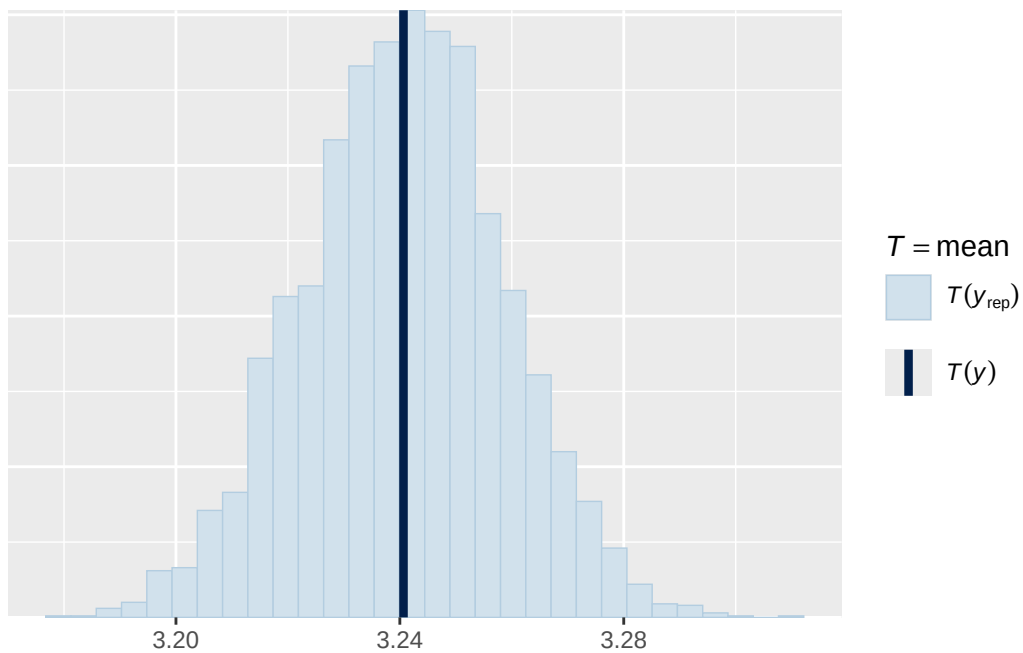
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

